

Mathematical Methods for Classical Mechanics

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Consider a particle moving along S^1 .

$$S^1 = \{(x, y) \mid x^2 + y^2 = 1\} \text{ is unit circle.}$$

Imagine $S^n = \left\{ (x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} x_i^2 = 1 \right\}$.

Claim 1. S^1 is compact.

Consider $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x^2 + y^2.$$

Observe f is continuous, and $f^{-1}\{1\} = S^1$ and hence S^1 will be closed and bounded. By Heine-Borel (in $\mathbb{R}^2$, we will see why this theorem holds for $\mathbb{R}^n$),

Proof. Hence, S^1 is compact. □

The parameterization of S^1 is given by the map

$$\gamma: \mathbb{R} \rightarrow S^1, \quad \theta \mapsto (\cos \theta, \sin \theta),$$

defined using Taylor series.

Exercise 1. Is there a map $\gamma: I \rightarrow S^1$ such that γ is invertible and differentiable (C^∞), i.e., smooth?

Consider the map

$$\gamma(\theta) = (\cos \theta, \sin \theta), \quad \theta \in [0, 2\pi].$$

This does not work as $[0, 2\pi]$ is compact, but S^1 is compact by the Heine-Borel theorem on $\mathbb{R}^2$.

Remark 0.1. We cannot use any interval as:

1. An open interval does not hold, in order to preserve continuity of the map.
2. A closed interval, derivative is defined one-sided for end points (not desirable).

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Let S^1 be the unit circle in $\mathbb{R}^2$ with the standard coordinate system.

Definition 0.1. Let P_N denote a stereographic projection from $N = (0, 1)$. The map is given by

$$P_N: S^1 \setminus \{N\} \rightarrow \mathbb{R},$$

$$(x, y) \mapsto \frac{x}{1 - y}.$$

Derivation of the formula. By the two-point formula, consider the line passing through $N = (0, 1)$ and a point $(x, y) \in S^1$. The x -intercept $(x_0, 0)$ of this line satisfies

$$\frac{y-1}{x-0} = \frac{1-0}{0-x_0}.$$

Since $(x, y) \in S^1$, we have $x^2 + y^2 = 1$. Solving for x_0 gives

$$x_0 = \frac{x}{1-y}.$$

□

Proposition 0.1. Check that P_N is bijective.

Proof. Consider the kernel of P_N :

$$\ker P_N = \left\{ \frac{x}{1-y} = 0 \right\}.$$

This implies $x = 0$ and $y \neq 1$. Since $(x, y) \in S^1$, we have $x^2 + y^2 = 1$. If $x = 0$, then $y = \pm 1$. However, $y = 1$ is excluded (as N is the projection point), so the only possible point is $(0, -1)$. But $(0, -1)$ maps to 0 under P_N , and no other point maps to 0 except $(0, -1)$. Thus, P_N is injective.

For surjectivity, given any $z \in \mathbb{R}$, we can find $(x, y) \in S^1$ such that $P_N(x, y) = z$. Solving for y in terms of x and z yields a unique point on S^1 for each $z \in \mathbb{R}$. Hence, P_N is bijective. □

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Remark 0.2. $\ker P_N = \{(0, -1)\}$;

Remark 0.3. P_N is injective $\iff \ker P_N = \{0\}$, but we are not getting $\{0\}$. Is $(0, -1)$ similar to “0”?

Surjective. We show that $P_N(S_1 \setminus N) = \mathbb{R}$.

For $\ker \mathbb{R}$,

- We show $\exists(\alpha_k, y_k) \in S_1 \setminus N$ such that

$$P_N(\alpha_k, y_k) = k$$

$$\frac{\alpha_k}{1-y_k} = k \implies \alpha_k = k(1-y_k)$$

Now, $\alpha_k^2 + y_k^2 = 1$,

$$k^2(1-y_k)^2 + y_k^2 = 1$$

and

$$k^2(1-2y_k+y_k^2) + y_k^2 = 1$$

$$(k^2+1)y_k^2 - 2ky_k + (k^2-1) = 0$$

$$\Delta_y = (4k)^2 - 4(k^2+1)(k^2-1)$$

$$\Delta_y = 16k^2 - 4k^4 - 4 \geq 0 \implies (\alpha_k, y_k) \text{ exist.}$$

□

$$P_N^{-1}: \mathbb{R} \rightarrow S_1 \setminus N$$

$$k \mapsto \left(\frac{2k}{1+k^2}, \frac{k^2-1}{k^2+1} \right)$$

□

Check construction, and derive.

Exercise: Verify the following: The stereographic projection from $(0, -1)$ and is given by

$$P_S: S' \setminus S \rightarrow \mathbb{R}$$

$$(\alpha, y) \mapsto \frac{\alpha}{1+y}$$

$$P_S^{-1}: \mathbb{R} \rightarrow S' \setminus S$$

$$t \mapsto \left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2} \right)$$

Same as P_N given above.

Definition 0.2 (Construction of P_N^{-1}). Join a line between $(t, 0)$ and $(0, 1)$:

$$\frac{x}{t} + \frac{y}{1} = 1$$

$$y = 1 - \frac{x}{t}$$

Substitute in $x^2 + y^2 = 1$, then we get

$$x^2 + \left(1 - \frac{x}{t}\right)^2 = 1$$

$$\implies x = \frac{2t}{1+t^2}, \quad y = \frac{1-t^2}{1+t^2} \quad (\text{Similar for } P_S)$$

Observe that P_S, P_N are rational functions and hence differentiable.

Definition 0.3. We call the map P_N (or P_S) a chart/coordinate system.

Definition 0.4. P_S and P_N considered together is called an atlas.

Remark 0.4. S^1 admits an atlas consisting of two stationary points P_S, P_N .

Proposition 0.2. The composition $P_S \circ P_N^{-1}: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$ is differentiable.

Proof. Exercise: Verify the fact given above. Composition of differentiable functions is differentiable. $\square$

Observe that $P_S \circ P_N^{-1}$ takes X to Y , then $(P_S \circ P_N^{-1})^{-1} = P_N \circ P_S^{-1}$ is the inverse of the “change of coordinates.”

1. **Exercise:** Generalize the above example to S^2 and find the expression for P_N , P_S , $P_N \circ P_S^{-1}$, $P_S \circ P_N$.

Definition 0.5. Consider the projection maps for $S^2 \setminus \{(0, 0, 1)\}$:

$$P_N : S^2 \setminus \{(0, 0, 1)\} \rightarrow \mathbb{R}^2, \quad (x, y, z) \mapsto \left(\frac{x}{1-z}, \frac{y}{1-z} \right).$$

We get the formula from similarity of triangles.

Definition 0.6. The inverse projection map is given by:

$$P_N^{-1} : \mathbb{R}^2 \rightarrow S^2 \setminus \{(0, 0, 1)\}, \quad (x, y) \mapsto \left(\frac{2x}{x^2 + y^2 + 1}, \frac{2y}{x^2 + y^2 + 1}, \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1} \right).$$

2. Work out the expression for the case of S^N :

$$P_N : S^N \setminus \{(0, \dots, 0, 1)\} \rightarrow \mathbb{R}^N, \quad (x_1, \dots, x_N, x_{N+1}) \mapsto \left(\frac{x_1}{1-x_{N+1}}, \dots, \frac{x_N}{1-x_{N+1}} \right).$$

Definition 0.7. Given a space X , $n \in \mathbb{N}$. A *chart* for X is a pair (U, φ) where U is an open subset of X and $\varphi : U \rightarrow A$ is a homeomorphism, where A is an open subset of $\mathbb{R}^n$.

Definition 0.8. For two charts (U, φ) and (V, ψ) are *compatible* if $\varphi \circ \psi^{-1}$ and $\psi \circ \varphi^{-1}$ are differentiable (and invertible), and $U \cap V \neq \emptyset$.

Definition 0.9. An *atlas* is a collection of compatible charts $\{(U_i, \varphi_i)\}_{i \in I}$ such that the union of U_i 's contains every point of X .

Remark 0.5. By smooth structure we mean a way of doing calculus.

Definition 0.10. A *smooth structure* on X is a maximal atlas (of compatible charts).

Consider the following projections:

$$\begin{aligned} U_+ : S' \cap \{(x, y) \mid y > 0\} &\rightarrow (-1, 1) \\ U_+(x, y) &= x \end{aligned}$$

$$\begin{aligned} U_- : S' \cap \{(x, y) \mid y < 0\} &\rightarrow (-1, 1) \\ U_-(x, y) &= x \end{aligned}$$

Remark 0.6. Observe that $S' \cap \{(x, y) \mid y > 0\}$ and $S' \cap \{(x, y) \mid y < 0\}$ are open subsets (in which topology).

$$R_+ : S' \cap \{(x, y) \mid x > 0\} \rightarrow (-1, 1)$$

$$R_+(x, y) = y$$

$$R_- : S' \cap \{(x, y) \mid x < 0\} \rightarrow (-1, 1)$$

$$R_-(x, y) = y$$

Remark 0.7. Observe that $S' \cap \{(x, y) \mid x > 0\}$ and $S' \cap \{(x, y) \mid x < 0\}$ are open subsets (in which topology).

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Exercise 2. Show U_+ , U_- , R_+ , R_- are compatible charts.

Given domains, U_+ , U_- , R_+ , R_- are invertible.

Definition 0.11 (Continuity of U_+). U_+ is continuous.

For $\varepsilon > 0$, $\exists \delta_\varepsilon > 0$ such that

$$|x - x_0| < \varepsilon \quad \text{whenever} \quad \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta_\varepsilon.$$

We choose $\delta_\varepsilon = \varepsilon$, and the above condition is satisfied.

$$\lim_{n \rightarrow 0} \frac{U_+(x + \frac{1}{n}, y) - U_+(x, y)}{\frac{1}{n}} \quad \text{exists, } U_+ \text{ is } q\text{-differentiable.}$$

Claim 2. U_+^{-1} is a continuous function.

$$U_+^{-1} : (-1, 1) \rightarrow S' \cap \{(x, y) \mid y > 0\}$$

$$x \mapsto (x, \sqrt{1 - x^2}).$$

For every $\varepsilon > 0$, $\exists \delta_\varepsilon > 0$ such that

$$\sqrt{(x - x_0)^2 + (\sqrt{1 - x^2} - \sqrt{1 - x_0^2})^2} < \varepsilon \quad \text{whenever} \quad |x - x_0| < \delta_\varepsilon.$$

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Claim 3. If $|x - x_0| < \delta$ then

$$\left| (1 - x_0^2)^{1/2} - (1 - x^2)^{1/2} \right| < M\delta$$

where $M = M(x_0, \delta) > 0$.

Consider $f : (-1, 1) \rightarrow \mathbb{R}$ defined by

$$f(x) = (1 - x^2)^{1/2}.$$

Observe that f is continuous on $[-1, 1]$ and differentiable on $(-1, 1)$.

Proof. By the Mean Value Theorem (MVT), consider $(x, x_0) \subseteq (-1, 1)$ such that $\exists c \in (x, x_0)$ with

$$|f(x) - f(x_0)| = |f'(c)| \cdot |x - x_0|.$$

We compute the derivative of f :

$$f'(x) = \frac{-x}{(1 - x^2)^{1/2}}.$$

Thus,

$$|f'(c)| = \frac{|c|}{(1 - c^2)^{1/2}}.$$

Since $c \in (x, x_0)$ and $|x - x_0| < \delta$, we have $|c - x_0| < \delta \implies |c| < (|x_0| + \delta)$.

Define

$$M(x_0, \delta) = \frac{|x_0| + \delta}{(1 - (|x_0| + \delta)^2)^{1/2}} = M.$$

Therefore,

$$|f(x) - f(x_0)| < M\delta.$$

□

Now we can show that U_+^{-1} is continuous.

Observe that $f_1(x) = x$, $f_2(x) = (1 - x^2)^{1/2}$ for $x \in (-1, 1)$ is differentiable $\implies U_+^{-1} \equiv (f_1, f_2)$ is also differentiable.

If we can show U_-, R_+, R_- are invertible and differentiable $\implies$ They are compatible.

Remark 0.8. Inverse of a differentiable function is differentiable.

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$ and f is invertible and $f'(f^{-1}(x)) \neq 0$.

$$f'(x) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \text{ exists.}$$

$$f'(x) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{f^{-1}(f(x)) - f^{-1}(f(x_0))}$$

$$\frac{1}{f'(f^{-1}(x_0))} = \lim_{x \rightarrow x_0} \frac{f^{-1}(f(x)) - f^{-1}(f(x_0))}{f(x) - f(x_0)}$$

As $x_0 \in \mathbb{R}$, f is bijective, $\forall x \in \mathbb{R}$ st $f(x) = y$ and $f(x_0) = y_0$, as f is continuous

$$\frac{1}{f'(f^{-1}(y_0))} = \lim_{y \rightarrow y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0}$$

- Exercise: $\{U_+, U_-, R_+, R_-\}$ forms an atlas.
- Exercise: Verify that U_+ is compatible with P_N and P_S .

- **Fact:** Maximal atlas exists (like using Zorn's lemma).
- Exercise (RA II): An open subset of $\mathbb{R}^n$ admits a smooth structure.

Definition 0.12. The space X with a smooth structure is called a smooth manifold.

- Exercise: The n -sphere S^n is a smooth manifold.

Definition 0.13. The dimension $\dim X$ is given by n .

Example 0.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}^n$ be a differentiable map f^n ,

$$f = (f_1, \dots, f_n).$$

Definition 0.14. Let $f = (f_1, f_2, \dots, f_n)$ be a differentiable function. The derivatives of f are given by

$$\dot{f}_i = \frac{df_i}{dt}.$$

Definition 0.15. Let $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function. The derivative of g is given by the gradient

$$\nabla g = \left(\frac{\partial g}{\partial x_1}, \dots, \frac{\partial g}{\partial x_n} \right).$$

Definition 0.16. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a differentiable function. The derivative of h is given by the Jacobian matrix

$$Dh = \begin{pmatrix} \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_m}{\partial x_1} & \cdots & \frac{\partial h_m}{\partial x_n} \end{pmatrix}_{m \times n} = \left(\frac{\partial h_i}{\partial x_j} \right)_{m \times n}.$$

Definition 0.17. Let $M_n(\mathbb{R})$ be the set of all $n \times n$ matrices with real entries, i.e.,

$$M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}.$$

There is a metric and topology defined on $M_n(\mathbb{R})$. Observe that $M_n(\mathbb{R})$ is a smooth manifold.

Definition 0.18. Let $GL_n(\mathbb{R})$ be the general linear group of degree n over $\mathbb{R}$, defined as

$$GL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det A \neq 0\}.$$

Note that $GL_n(\mathbb{R})$ is an open subset of $M_n(\mathbb{R})$.

Argument (1). $GL_n(\mathbb{R})$ is open. If $A \in GL_n(\mathbb{R})$, then all sufficiently nearby matrices to A are also in $GL_n(\mathbb{R})$. $\square$

Test case, $n = 1$

$$GL(1, \mathbb{R}) = \{a_{11} \mid a_{11} \neq 0\} = \mathbb{R} - \{0\}$$

In general case, we can find $\delta > 0$ such that for any matrix $B \in M_n(\mathbb{R})$ within δ -distance of A , we have $\det(B) \neq 0$.

Exercise 3. Write down the proof.

Consider $A \in M_{n+1}(\mathbb{R})$ and $\det A \neq 0$. We know $M_{n+1}(\mathbb{R}) \cong \mathbb{R}^{(n+1)^2}$ and

$$A \mapsto \xi(A)$$

and $g : \mathbb{R}^{(n+1)^2} \xrightarrow{\det} \mathbb{R}$ which is a polynomial equation. So it is continuous.

$$\implies M_n(\mathbb{R}) \xrightarrow{\det} \mathbb{R} \text{ is continuous}$$

$$\implies \det^{-1}\{0\} \text{ will be a closed set and}$$

$$\implies M_n(\mathbb{R}) \setminus (\det^{-1}\{0\}) = \{A \mid \det A \neq 0\}$$

Note that $\det$ is polynomial f^n in terms of entries of input variable (= matrix).

1. Argument (2)

We show that $GL_n(\mathbb{C})^c$ is closed under taking limits

Definition 0.19. If $\{A_n\}$ is a sequence of matrices in $GL_n(\mathbb{C})^c$ that converges then $A \in GL_n(\mathbb{C})^c$.

Fact 1. The set $O_n(\mathbb{R}) = \{A \in M_n(\mathbb{C}) \mid AA^T = A^T A = I\}$

Where I is the identity matrix of order $n \times n$.

The content in the image corresponds to a definition and an exercise. The set $SO_n(\mathbb{R})$ is defined as follows:

$$SO_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid A^T A = AA^T = I \text{ and } \det A = 1\}$$

Definition 0.20. The special orthogonal group $SO_n(\mathbb{R})$ is the set of all $n \times n$ real matrices A such that:

1. $A^T A = AA^T = I$ (i.e., A is orthogonal),
2. $\det A = 1$.

Exercise 4. Show that if $A \in O_n(\mathbb{R})$ and $\det(A) \neq 1$, then $\det(A) = -1$.

Proof. Since $A \in O_n(\mathbb{R})$, we have $A^T A = I$. Taking determinants on both sides, we get:

$$\det(A^T A) = \det(I).$$

Using the property of determinants, $\det(A^T) = \det(A)$ and $\det(I) = 1$, we have:

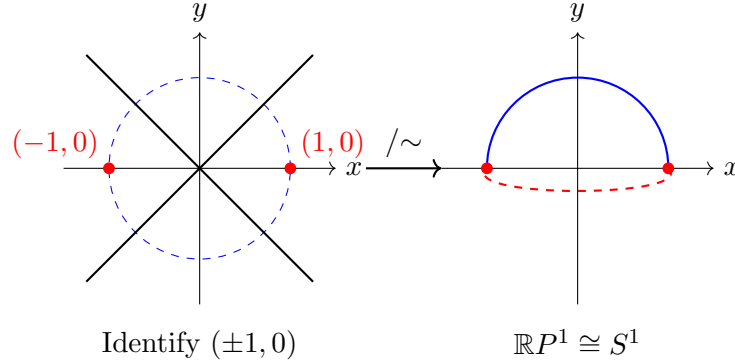
$$\det(A^T) \det(A) = 1 \implies (\det(A))^2 = 1.$$

Thus, $\det(A) = \pm 1$. Given that $\det(A) \neq 1$, it follows that $\det(A) = -1$. □

Definition 0.21. The real projective space of dimension 1

The space of lines that pass through the origin in $\mathbb{R}^2$.

Remark 0.9. Identifying the points $\{(\pm 1, 0)\}$ to a single point gives us the space of lines passing through the origin.

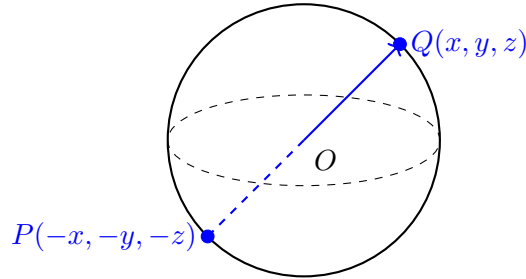


This identification is equivalent to the real projective space $\mathbb{R}P^1$ and can be visualized as a circle, where opposite points are identified.

(Look at **Quotient Topology**)

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The space of lines in $\mathbb{R}^3$ can be visualized as follows:



Definition 0.22. The **real projective space of dimension n** is the space of lines passing through the origin in $\mathbb{R}^{n+1}$. It is denoted by $\mathbb{R}P^n$.

$$\mathbb{R}P^n = \frac{\mathbb{R}^{n+1} \setminus \{0\}}{\sim}$$

where $(v = \lambda v', \lambda \neq 0)$ defines the equivalence relation $\sim$.

$$\mathbb{R}P^n = \{[\alpha_1, \alpha_2, \dots, \alpha_n, \alpha_{n+1}] \mid \text{at least one } \alpha_i \text{ is non-zero}\}$$

This represents the line passing through $(\alpha_1, \dots, \alpha_{n+1})$.

Remark 0.10. Observe that $[\alpha_1, \alpha_2, \dots, \alpha_n, \alpha_{n+1}] = [y_1, \dots, y_n, y_{n+1}]$ if and only if there exists $\lambda \neq 0$ such that $\alpha_i = \lambda y_i$ for each $i = 1, \dots, n+1$.

Exercise: $\mathbb{R}P^2$ is not homeomorphic to S^2 .

Definition 0.23. $\mathbb{R}P^2$ is non-orientable, while S^2 is orientable.

Theorem 0.1. $\mathbb{R}P^3 \cong SO(3)$, i.e. homeomorphism.

Definition 0.24. $\mathbb{R}P^3 = \frac{\mathbb{R}^4 \setminus \{0\}}{\sim}$ where $v \sim \lambda v$ for $\lambda \neq 0$.

$$\mathbb{R}P^3 = \{[x_1, x_2, x_3, x_4] \mid \text{at least one } x_i \text{ is not zero}\}$$

Definition 0.25. $SO_3(\mathbb{R}) = \{\text{All 3-dimensional orthogonal matrices with determinant} = 1\}$

$$SO_3(\mathbb{R}) = \{\text{All rotation matrices in } \mathbb{R}^3\}$$

Observe that for any line passing through the origin, we can represent it by the following:

$$[\alpha_1, \dots, \underline{\alpha_i}, \dots, \alpha_{n+1}] = \left[\frac{\alpha_1}{\alpha_i}, \dots, 1, \frac{\alpha_{i+1}}{\alpha_i}, \dots, \frac{\alpha_{n+1}}{\alpha_i} \right]$$

We can represent a line by

$$[y_1, y_2, \dots, 1, \dots, y_{n+1}]$$

We will show that $\mathbb{R}P^n$ is a smooth manifold.

$$\varphi_1: \mathbb{R}P^n \rightarrow \mathbb{R}^n$$

$$\varphi_1: \{[\alpha_1, \dots, \alpha_{n+1}] \mid \alpha_i \neq 0\} \rightarrow (y_2, \dots, y_{n+1})$$

We can form a chart in $\mathbb{R}P^1$ space.

$$\varphi_1: \{[\alpha, y] \mid \alpha \neq 0\} \rightarrow \text{removes } y\text{-axis}$$

$$\varphi_2: \{[\alpha, y] \mid y \neq 0\} \rightarrow \text{removes } x\text{-axis}$$

Now in $\mathbb{R}P^n$ we can define

$$\varphi_i: \{[\alpha_1, \dots, \underline{\alpha_i}, \dots, \alpha_{n+1}] \mid \alpha_i \neq 0\} \rightarrow \mathbb{R}^n$$

$$[\alpha_1, \dots, \alpha_{n+1}] \mapsto \left(\frac{\alpha_1}{\alpha_i}, \dots, \frac{\alpha_{i-1}}{\alpha_i}, \frac{\alpha_{i+1}}{\alpha_i}, \dots, \frac{\alpha_{n+1}}{\alpha_i} \right)$$

Claim 4. The collection of the above charts forms an atlas. (Prove): We have to show compatibility.

- The domains have a non-empty intersection.
- Both are differentiable, invertible maps and as we work in finite dim, φ_i will have a continuous inverse. Hence, all φ_i are compatible. (✓)

In the case of $\mathbb{RP}^2$,

$\varphi_2 \circ \varphi_1^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is differentiable map in the usual sense.

Definition 0.26.

$$\mathbb{RP}^n = S^n / \sim$$

where $a \sim b$ if $a = b$ or $a = -x, b = -x$ or $a = -x, b = x$.

Let X be a smooth manifold. X has an atlas $\mathcal{A}$.

Definition 0.27. Let $f : X \rightarrow \mathbb{R}$ be f^n . Let $x_0 \in X$ be a point. We say f is differentiable at x_0 if for any chart (U, φ) containing x_0 , the function $f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}$ is differentiable (smooth) at $\varphi(x_0)$.

Definition 0.28. Let (a, b) denote an open interval in $\mathbb{R}$. We say that $g : (a, b) \rightarrow X$ is differentiable, if for a chart (V, ψ) , the function

$$\psi \circ g : (a, b) \rightarrow \mathbb{R}^n$$

is differentiable at all points in (a, b) .

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Definition 0.29. Using Definition ??, we can define differentiability of function

$$g : X \rightarrow \mathbb{R}^m.$$

We say that $g : X \rightarrow \mathbb{R}^m$ is differentiable (diff.) if each component of g , say $g_i : X \rightarrow \mathbb{R}$ (where $g_i = \pi_i \circ g$) is differentiable. Say $g = (g_1, \dots, g_m)$.

Definition 0.30. In general, if $X \rightarrow Y$ are smooth manifolds and $f : X \rightarrow Y$ be a function, we say that f is differentiable (diff.) or smooth if for any chart (U, φ) and (V, ψ) , where (U, φ) is a chart for X , and (V, ψ) is a chart for Y ,

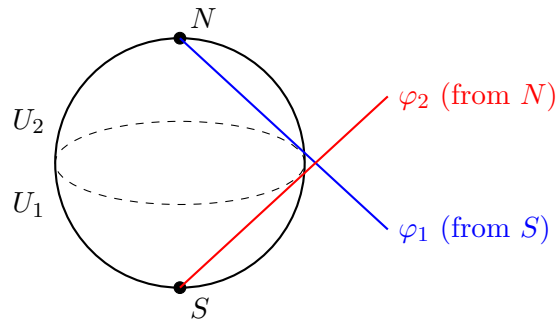
$$\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \rightarrow \psi(V)$$

is differentiable as a function $\mathbb{R}^n \rightarrow \mathbb{R}^m$ and check that $\text{Im}(f) \cap V \neq \emptyset$.

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Example 0.2. 1. Euclidean space $\mathbb{R}^n$ is a manifold, with an atlas of one chart.

$$2. S^2 = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$$



S^2 with two stereographic charts

3) For a planar pendulum, the configuration space is given by the map

$$\varphi: \mathbb{R} \rightarrow S^1, \quad t \mapsto (\cos t, \sin t)$$

with charts $U_1 = (-\pi, \pi)$ and $U_2 = (0, 2\pi)$.

4)

1. Double Pendulum $\Rightarrow T^2 = S^1 \times S^1$

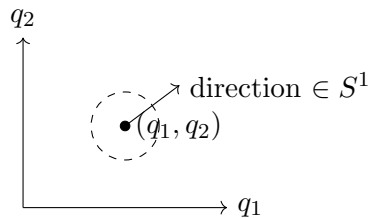
2. Spherical Pendulum $\Rightarrow S^2$

3. Double Spherical Pendulum $\Rightarrow S^2 \times S^2$

5) We get coordinate system

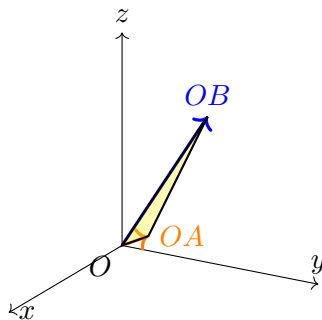
$$\mathbb{R}^2 \times S^1$$

for a line in the plane.



Position $(q_1, q_2) \in \mathbb{R}^2$, direction $\in S^1$

6) Given: Rigid right $\triangle OAB$ moves around O and is given by vector $OA \in S^2$ and $OB \in S^1$ will be a vector on the plane $\perp^{ever}$ to OA .



The three vectors we get are

$$\mathbf{e}_1 = \frac{OA}{|OA|}, \quad \mathbf{e}_2 = \frac{OB}{|OB|},$$

and $\mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2$. Equivalently, to

$$M = \begin{pmatrix} 1 & 1 & 1 \\ \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \end{pmatrix}$$

with $\det M = 1$.

Problem 1. $SO(3)$ is homeomorphic to $\mathbb{RP}^3$.

Definition 0.31. The dimension of configuration space is $\leq$ degree of freedom.

Definition 0.32. What are embedded manifolds? How to show every manifold can be embedded in a Euclidean space?

Let X be a smooth manifold and p be a point in X . The tangent space to X at p is of dimension n .

There is a chart for X , say (U, φ) , containing point p . Assume $p = (0, \dots, 0)$ (an n -tuple).

—

Definition 0.33. Consider tangent vectors at a point p as velocity vectors.

We obtain curves passing through point p by taking inverse images of straight line segments under φ .

Proof. We get curves $\gamma_v(t) = \varphi^{-1}(ct \cdot v)$, where $t \in (-\varepsilon, \varepsilon)$ and $\gamma_v(0) = \varphi^{-1}(0) = p$. □

—

—

We say that two locally defined curves γ_1 and γ_2 through $(= \gamma_1(0) = \gamma_2(0) = p)$ are **equivalent** for any chart if

$$\left. \frac{d}{dt}(\varphi \circ \gamma_1) \right|_{t=0} = \left. \frac{d}{dt}(\varphi \circ \gamma_2) \right|_{t=0}.$$

Definition 0.34. Two locally defined curves γ_1 and γ_2 through a point p are called **equivalent** if for any chart φ around p ,

$$\left. \frac{d}{dt}(\varphi \circ \gamma_1) \right|_{t=0} = \left. \frac{d}{dt}(\varphi \circ \gamma_2) \right|_{t=0}.$$

Remark 0.11. Observe that this can construct an equivalence relation on the set of curves that pass through p .

Independence of the choice of chart. The independence of the definition of the choice of chart is shown as follows:

$$\left. \frac{d}{dt}(\varphi \circ \gamma_1) \right|_{t=0} = \left. \frac{d}{dt}(\varphi \circ \gamma_2) \right|_{t=0}$$

implies

$$\left. \frac{d}{dt}(\psi \circ \varphi^{-1} \circ \varphi \circ \gamma_1) \right|_{t=0} = \left. \frac{d}{dt}(\psi \circ \varphi^{-1} \circ \varphi \circ \gamma_2) \right|_{t=0}.$$

Applying the chain rule,

$$D(\psi \circ \varphi^{-1})(\varphi \circ \gamma_1) \cdot \left. \frac{d}{dt}(\varphi \circ \gamma_1) \right|_{t=0} = D(\psi \circ \varphi^{-1})(\varphi \circ \gamma_2) \cdot \left. \frac{d}{dt}(\varphi \circ \gamma_2) \right|_{t=0}.$$

□

As $\varphi_0 r_1(0) = \varphi_0 r_2(0) = 0$ we get,

$$D(\varphi_0 \varphi^{-1}) \cdot \left. \frac{d}{dt}(\varphi_0 r_1) \right|_{t=0} = D(\varphi_0 \varphi^{-1}) \cdot \left. \frac{d}{dt}(\varphi_0 r_2) \right|_{t=0}$$

Remark 0.12. Observe how the atlas structure makes the equivalence relation become independent of the chart.

Definition 0.35. **Tangent vector** to X at p is an equivalence class of curves that pass through p .

Remark 0.13. For computational purposes, consider the straight line segment in a chart.

Definition 0.36. The set of all tangent vectors at p is called a **tangent space** to X at p . Denoted by $T_p X$.

Remark 0.14. In a local coordinate system, if $x \in T_p X$ then

$$x = \sum_{i=1}^n c_i \frac{\partial}{\partial x_i}$$

where $\frac{\partial}{\partial x_i}$ denotes the tangent vector along the i th coordinate of the chart.

Tangent vector as derivations

Recall

If V is a vector space over $\mathbb{R}$ then we have $T : V \rightarrow \mathbb{R}$ linear functional.

Observe, $V^* = \{T : V \rightarrow \mathbb{R}\}$ is a vector space.

Consider $C^\infty(\mathbb{R}^n) = \{f \mid f : \mathbb{R}^n \rightarrow \mathbb{R}, f \text{ is smooth}\}$.

Exercise 5. $C^\infty(\mathbb{R}^n)$ is a vector space over $\mathbb{R}$. (Note: Hard Check)

Remark 0.15. $C^\infty(\mathbb{R}^n)$ is more than a vector space, if $f, g \in C^\infty(\mathbb{R}^n)$ then $f(x)g(x)$. This is an algebra.

Definition 0.37. A derivation on $C^\infty(\mathbb{R}^n)$ is a linear functional

$$v : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$$

such that

$$v(fg) = v(f)g + fv(g) \quad (\text{Leibniz Rule}).$$

Example 0.3.

$$v = \left. \frac{\partial}{\partial x_i} \right|_a \quad \text{where } a \in \mathbb{R}^n$$

Proof. $\left. \frac{\partial}{\partial x_i} \right|_a$ is a linear functional, $v : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$ and

$$v(fg) = v(f)g + fv(g).$$

□

$$\left. \frac{\partial}{\partial x_i} \right|_a (cf) = c \left. \frac{\partial f}{\partial x_i} \right|_a$$

$$\left. \frac{\partial}{\partial x_i} \right|_a (f + g) = \left. \frac{\partial f}{\partial x_i} \right|_a + \left. \frac{\partial g}{\partial x_i} \right|_a$$

$$\left. \frac{\partial}{\partial x_i} \right|_a (fg) = f(c_a) \left. \frac{\partial g}{\partial x_i} \right|_a + \left. \frac{\partial f}{\partial x_i} \right|_a g(c_a)$$

Definition 0.38. If u is a vector in $\mathbb{R}^n$, then the directional derivatives

$$D_u : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$$

are defined as

$$D_u(f) \Big|_a = D_u f = \langle \nabla f|_a, u \rangle.$$

Exercise:

1. Set of all derivations forms a vector space
- 2.

Definition 0.39. Show that the set $\left\{ \left. \frac{\partial}{\partial x_1} \right|_a, \left. \frac{\partial}{\partial x_2} \right|_a, \dots, \left. \frac{\partial}{\partial x_n} \right|_a \right\}$ forms a basis for $\mathcal{D}(\mathbb{R}^n)$.

For $v \in \mathcal{D}(\mathbb{R}^n)$,

$$v = \sum_{i=1}^n c_i \left. \frac{\partial}{\partial x_i} \right|_a, \quad \text{where } c_i \in \mathbb{R}.$$

Set,

$$c_j := v(\pi_j),$$

where π_j is the projection map onto the j -th coordinate.

$$v(f) := \left(\sum_{i=1}^n c_i \left. \frac{\partial}{\partial x_i} \right|_a \right) (f).$$

Proof. Observe that

$$\left\{ \frac{\partial}{\partial x_i} \right\}_{i=1}^n$$

is a linearly independent set of size n . Therefore, it is a basis of the space of derivations. $\square$

Let v denote a tangent vector (an equivalence class of curves through p). In particular, $v = [\gamma]$ where γ is a representative of v .

We can think of v as a derivation as follows:

$$v : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}, \quad v(f) = \left. \frac{d}{dt}(f \circ \gamma) \right|_{t=0}$$

Exercise 6. Show v gives a well-defined derivation.

Proof. If $v = [\gamma_1] = [\gamma_2]$; $\left. \frac{d}{dt}\gamma_1 \right|_{t=0} = \left. \frac{d}{dt}\gamma_2 \right|_{t=0}$,

$$\left. \frac{d}{dt}(f \circ \gamma_1) \right|_{t=0} = \left. \frac{d}{dt}(f \circ \gamma_2) \right|_{t=0}$$

By chain rule,

$$Df|_{\gamma_1(0)} \circ \left. \frac{d}{dt}\gamma_1 \right|_{t=0} = Df|_{\gamma_2(0)} \circ \left. \frac{d}{dt}\gamma_2 \right|_{t=0}$$

$\square$

Let M be a smooth manifold of dimension n .

The **tangent bundle** is given by

$$TM = \bigcup_{p \in M} T_p M = \{(p, v) \mid v \in T_p M \text{ and } p \in M\}.$$

Definition 0.40. Claim: TM is a smooth manifold, it admits an atlas $\{(V_i, \varphi_i)\}_{i \in I}$, we give an atlas on TM .

For each $p \in M$, the tangent space $T_p M$ is defined as

$$T_p M = \left\langle \left. \frac{\partial}{\partial x_1} \right|_p, \left. \frac{\partial}{\partial x_2} \right|_p, \dots, \left. \frac{\partial}{\partial x_n} \right|_p \right\rangle.$$

Every $v \in T_p M$ has a unique representation by $(v_1, v_2, \dots, v_n)$, where

$$v = \sum_{i=1}^n v_i \left. \frac{\partial}{\partial x_i} \right|_p.$$

Thus, $(x_1, \dots, x_n, v_1, \dots, v_n)$ represent (p, v) .

—

Thus, we define a chart as follows:

Definition 0.41. Let $V_i = \{(p, v) \mid v \in T_p M \text{ and } p \in V_i\}$, where V_i is an open cover of M .

Define the chart $\Psi_i: V_i \rightarrow \mathbb{R}^n \times \mathbb{R}^n \cong \mathbb{R}^{2n}$ by

$$\Psi_i(p, v) = (\varphi(p), v_1, v_2, \dots, v_n),$$

where $\varphi: V_i \rightarrow \mathbb{R}^n$ is a local coordinate map and $v = \sum_{i=1}^n v_i \frac{\partial}{\partial x_i} \Big|_{\varphi(p)}$.

To verify the claim, we need to check the following:

1. $TM = \bigcup_i V_i$ (Exercise).
2. The transition functions are smooth.

Remark 0.16. Observe $\Psi_i \circ \Psi_j^{-1}$:

$$\Psi_j(V_i \cap V_j) \xrightarrow{\Psi_i \circ \Psi_j^{-1}} \Psi_i(V_i \cap V_j).$$

—

—

$$\varphi_j \circ \varphi_i^{-1} [(\alpha, u) \mapsto (\varphi_j \circ \varphi_i^{-1}(\alpha), d(\varphi_j \circ \varphi_i^{-1})(u))]$$

$$d(\varphi_j \circ \varphi_i^{-1})$$

Definition 0.42. The above composition is smooth. Note the vector $\alpha \in T_\alpha \mathbb{R}^{2n}$ is by applying the Jacobian of $\varphi_j \circ \varphi_i^{-1}$ at α .

Proof. The smoothness of the composition $\varphi_j \circ \varphi_i^{-1}$ follows from the fact that both φ_i and φ_j are smooth diffeomorphisms, and the composition of smooth maps is smooth. The Jacobian of $\varphi_j \circ \varphi_i^{-1}$ at α is well-defined and smooth, ensuring the tangent vector transformation is valid. $\square$

—

Example 0.4. 1. The tangent bundle of $\mathbb{R}^n$ is given by

$$T\mathbb{R}^n = \{(p, v) \mid v \in T_p \mathbb{R}^n, p \in \mathbb{R}^n\} = \mathbb{R}^n \times \mathbb{R}^n.$$

2. Let $M = S^1$ (the circle group). The tangent bundle of S^1 is

$$TS^1 = \{(p, v) \mid v \in T_p S^1, p \in S^1\} \cong S^1 \times \mathbb{R}$$

(diffeomorphic).

3. Let $M = S^2$. The tangent bundle of S^2 is

$$TS^2 = \{(p, v) \mid p \in S^2, v \in T_p S^2\}.$$

The guess that $TS^2 \cong S^2 \times \mathbb{R}^2$ is **not true**.

Consider the map $\pi : TM \rightarrow M$ defined by

$$\pi(p, v) = p.$$

Claim 5. The map π is a smooth map. That is, $\varphi \circ \pi \circ \varphi^{-1}$ is differentiable.

Proof. To verify the claim, suppose M and N are two smooth manifolds and

$$\varphi : M \rightarrow N$$

is a smooth map. Define the differential of φ by

$$D\varphi(p, v) = (\varphi(p), d\varphi(v)).$$

The map $D\varphi$ is smooth and linear when restricted to each tangent space. □

Definition 0.43. A **vector field** is a smooth function $s : M \rightarrow TM$ such that $\pi \circ s = \text{id}_M$.

Definition 0.44. The **flow generated by a vector field** is a smooth 1-parameter family $\phi_t : M \rightarrow M$ with $\phi_0 = \text{id}_M$. It is called the flow generated by vector field X on M .

Remark 0.17. Fact: Each vector field on a compact manifold admits a flow.

Example 0.5. Consider the vector field $X : S^2 \rightarrow TS^2$ defined by

$$X = \{(p, v) \mid \|p\| = 1, \langle p, v \rangle = 0\}.$$

Here, $S^2 \subseteq \mathbb{R}^3 \times \mathbb{R}^3$ and $p = (x, y, z)$ with $x^2 + y^2 + z^2 = 1$. Let

$$P = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\},$$

$$Q = \{(u, v, w) \mid \langle (x, y, z), (u, v, w) \rangle = 0\}.$$

Then $P \cap Q$ is 1-dimensional. Take the unit vector in $P \cap Q$ based at (x_0, y_0, z_0) . Scale it by a factor $(1 - z_0)(1 + z_0)$.

—

Example 0.6. 1-forms and co-tangent bundles

On $(\mathbb{R}^n, (x_1, \dots, x_n))$ we have $dx_i : \mathbb{R}^n \rightarrow \mathbb{R}$

The linear map is given by

$$dx_i(v) = c_i \quad \text{where} \quad v = \sum_{i=1}^n c_i e_i$$

and $e_i = (0, \dots, 1, \dots, 0)$ with 1 in the i^{th} place.

Exercise 7. Verify dx_i is a linear map. (Trivial)

By definition $dx_i \in (\mathbb{R}^n)^* = \{f : \mathbb{R}^n \rightarrow \mathbb{R} \mid f \text{ is linear}\}$.

Note that $(\mathbb{R}^n)^*$ is a vector space of dimension n with a basis $\{dx_1, \dots, dx_n\}$.

We can repeat the above argument for each tangent space $p \in \mathbb{R}^n$:

$$T_p \mathbb{R}^n = \left\langle \frac{\partial}{\partial x_1} \Big|_p, \dots, \frac{\partial}{\partial x_n} \Big|_p \right\rangle$$

We consider the dual space $(T_p \mathbb{R}^n)^*$.

Definition 0.45. The $(T_p \mathbb{R}^n)^*$ is the cotangent space at p (or the space of co-vectors).

Thus, we have $dx_i : T_p \mathbb{R}^n \rightarrow \mathbb{R}$,

$$v = \sum_{j=1}^n c_j \frac{\partial}{\partial x_j} \Big|_p \longmapsto c_i.$$

When we drop reference to point p , we get a family of functionals on the corresponding tangent space:

$$dx_i : T_p \mathbb{R}^n \rightarrow \mathbb{R}.$$

When we drop reference to point p , we get a family of functionals on corresponding tangent spaces:

$$dx_i : \mathcal{X}(\mathbb{R}^n) \rightarrow C^\infty(\mathbb{R}^n),$$

where $\mathcal{X}(\mathbb{R}^n)$ is the space of smooth vector fields.

Proof. For a vector field $v \in \mathcal{X}(\mathbb{R}^n)$, we have

$$dx_i(v) = f(p) = c_i,$$

where $v = \sum_{j=1}^n c_j \frac{\partial}{\partial x_j}$ and f is a smooth function such that $v(f) = c_i$ at p .

Thus, dx_i is a linear map from $\mathcal{X}(\mathbb{R}^n)$ to $C^\infty(\mathbb{R}^n)$. □

If we have a smooth function $f \in C^\infty(\mathbb{R}^n)$ and vector field $v \in \mathfrak{X}(\mathbb{R}^n)$, the function-valued vector field fv is a smooth vector field given by

$$(fv)(p) = f(p) \cdot v(p)$$

Definition 0.46. $\mathfrak{X}(\mathbb{R}^n)$ is a module over $C^\infty(\mathbb{R}^n)$.

Definition 0.47. Let X be a smooth manifold of dimension n . The cotangent bundle of X , as a set, is given by

$$T^*X = \bigcup_{p \in X} (T_p X)^* = \{(p, v) \mid p \in X, v \in (T_p X)^*\}.$$

In the case of $\mathbb{R}^n$, the cotangent bundle is

$$T^*\mathbb{R}^n = \bigcup_{p \in \mathbb{R}^n} (T_p \mathbb{R}^n)^*.$$

Suppose $(\alpha_1, \dots, \alpha_n)$ is a coordinate system on $\mathbb{R}^n$. Then the cotangent bundle can be expressed as

$$T^*\mathbb{R}^n = \{(p, \xi) \mid p \in \mathbb{R}^n, \xi \in (T_p \mathbb{R}^n)^*\} = \{(x_1, \dots, x_n, c_1, \dots, c_n) \mid \xi = \sum_{j=1}^n c_j d\alpha_j\} \cong \mathbb{R}^n \times \mathbb{R}^n \quad (\text{Bundle isomorphism})$$

Claim 6. The cotangent bundle T^*X is a smooth manifold of dimension $2n$.

Exercise 8. Set up an atlas for T^*X .

Check. Consider a chart on T^*X :

$$\varphi_\alpha: T^*X \rightarrow \mathbb{R}^{2n}, \quad (p, \xi) \mapsto (\varphi(p), \xi(\frac{\partial}{\partial x_1}), \dots, \xi(\frac{\partial}{\partial x_n})).$$

Here, $\xi = \sum_{i=1}^n \xi_i dx_i$ with $\xi_i \in \mathbb{R}$ for $1 \leq i \leq n$, and $dx_i(\frac{\partial}{\partial x_j}) = \delta_{ij}$. □

Let X be a smooth manifold of dimension n . Then, the cotangent bundle

$$T^*X = \bigcup_{p \in X} (T_p X)^*$$

is a smooth manifold of dimension $2n$.

$$T^*X = \{(p, v) \mid v \in (T_p X)^*, p \in X\}.$$

Example 0.7. 1. $T^*\mathbb{R}^n = \mathbb{R}^n \times (\mathbb{R}^n)^* \cong \mathbb{R}^n \times \mathbb{R}^n$

$$2. T^*S^1 = \{(p, v) \mid v \in (T_p S^1)^*, p \in S^1\}$$

Consider $T_p S^1 = \{\alpha: T_p S^1 \rightarrow \mathbb{R} \mid v \mapsto \alpha(v)\}$.

By the Riesz Representation Theorem, each $\alpha \in (T_p S^1)^*$ corresponds to a unique vector w such that

$$\alpha(v) = \langle w, v \rangle.$$

Let e be a fixed unit vector in $T_p S^1$. Then, for any $v \in T_p S^1$, we have

$$\alpha(v) = \langle w, v \rangle = \lambda \langle e, v \rangle = \lambda \|v\|,$$

where $\lambda = \frac{\alpha(v)}{\|v\|}$.

A 1-form on X is a smooth section of the cotangent bundle T^*X .

$$T^*X = \bigsqcup_{(p,\alpha)} \{ \alpha \in (T_p X)^*, p \in X \}$$

Therefore for any 1-form β , we have $\pi \circ \beta = \text{id}_X$. In other words, β is a smooth assignment of covectors to points in X .

Example 0.8. 1. Let $X = \mathbb{R}$. Then $T^*\mathbb{R} \cong \mathbb{R} \times \mathbb{R}^*$,

$$T^*\mathbb{R} = \bigsqcup_{(x,\alpha)} \{ \alpha \in (T_x \mathbb{R})^*, x \in \mathbb{R} \}.$$

Define $\beta : \mathbb{R} \rightarrow T^*\mathbb{R}$ by

$$x \mapsto (x, dx),$$

where

$$dx = \begin{cases} \alpha & \text{if } x = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Take the map $B_1 : \mathbb{R} \rightarrow T^*\mathbb{R}$ defined by

$$x \mapsto (x, x \, da).$$

Then B_1 is a 1-form.

1. For $X = \mathbb{R}^n$, to each 1-form B on $\mathbb{R}^n$, we have the following

$$B(x_1, \dots, x_n) = \sum_{i=1}^n c_i(x_1, \dots, x_n) da_i,$$

where c_i 's are smooth functions on $\mathbb{R}^n$.

Definition 0.48. The map $B : \mathbb{R}^n \rightarrow (T^*\mathbb{R}^n)$ is defined by

$$x \mapsto \left(x, \sum_{i=1}^n c_i(x) da_i \right).$$

2. Verify that B is a 1-form.

$$\begin{aligned} B : \mathbb{R}^3 &\rightarrow T^*\mathbb{R}^3, \\ (x, y, z) &\mapsto (x, y, z, dz - y \, dx). \end{aligned}$$

Proof. Check linearity. Do we just have to check linearity? $\pi \circ B = \text{id}$. □

1. Take $X = S^1 \subset \mathbb{R}^2$,

$$S^1 = \{(x, y) \mid x^2 + y^2 = 1\}.$$

We define $B_{(x,y)} = x \, dy$.

Definition 0.49. The 1-form B on S^1 is defined as $B_{(x,y)} = x dy$.

2. Take $X = S^3 \subset \mathbb{R}^4$,

$$S^3 = \{x \in \mathbb{R}^4 \mid \|x\| = 1\}.$$

Define

$$B_{(x_1, x_2, x_3, x_4)} = x_1 dx_1 + x_2 dx_2 + x_3 dx_3 + x_4 dx_4.$$

Then B is a 1-form on S^3 .

Proof. To verify that B is a 1-form on S^3 , we check two things:

- (a) Linearity of B on tangent vectors.
- (b) The condition $\pi^* \circ B = \text{id}_X$, where π^* is the pullback of B to the ambient space.

Remark 0.18. We do not just check linearity; we also check that $\pi^* \circ B = \text{id}_X$.

□

Let $(\alpha_1, \dots, \alpha_n)$ be the coordinate system on $\mathbb{R}^n$.

Definition 0.50. An exterior 2-form is a bilinear map that is skew-symmetric on $\mathbb{R}^n \times \mathbb{R}^n$ and real-valued.

In particular, if ω is an exterior 2-form (or 2-form) by definition

$$\omega: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R},$$

then

- 1. ω is linear in both variables,
- 2. ω is skew-symmetric, i.e.,

$$\omega(u, v) = -\omega(v, u) \quad \text{for all } u, v \in \mathbb{R}^n.$$

Example 0.9. Consider the map

$$\omega: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R},$$

defined by

$$(u, v) \mapsto \det \begin{pmatrix} u_1 & u_2 \\ v_1 & v_2 \end{pmatrix},$$

where $u = (u_1, u_2)$ and $v = (v_1, v_2)$. Observe that $\omega(u, v)$ gives the area of the parallelogram formed by u and v .

Exercise

Let $w: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ be an exterior 2-form such that $w \equiv 0$ if $u = v$. Show that

$$w(\lambda u, v) = \lambda w(u, v) \quad \text{where } \lambda, v \in \mathbb{R}^n,$$

and

$$w(u, u) = -w(u, u) \implies w(u, u) = 0 \quad \text{for any } u \in \mathbb{R}^n \text{ and } u \neq 0.$$

Definition 0.51. Show that the following is an exterior 2-form on $\mathbb{R}^3$. Fix $v \neq 0 \in \mathbb{R}^3$. Define

$$w(u_1, u_2) = v \cdot (u_1 \times u_2).$$

Example 0.10. Let α_1 and α_2 be two covectors on $\mathbb{R}^4$. Then we construct the following map:

$$s(u, v) = \begin{vmatrix} \alpha_1(u) & \alpha_2(u) \\ \alpha_1(v) & \alpha_2(v) \end{vmatrix}.$$

Check if s is a 2-form.

We denote $s = \alpha_1 \wedge \alpha_2$.

Exercise 9. Take any 2-form ω on $\mathbb{R}^n$, show that

$$\omega = \sum_{i < j} c_{ij} dx_i \wedge dx_j$$

for scalars c_{ij} . For $u, v \in T_p \mathbb{R}^n$,

$$\begin{aligned} \omega(u, v) &= \sum_{i < j} c_{ij} dx_i \wedge dx_j(u, v) \\ &= \sum_{i < j} c_{ij} \begin{vmatrix} dx_i(u) & dx_i(v) \\ dx_j(u) & dx_j(v) \end{vmatrix} = \begin{vmatrix} u_1 & v_1 \\ u_2 & v_2 \end{vmatrix} \\ &= - \sum_{i < j} c_{ij} \begin{vmatrix} dx_i(v) & dx_i(u) \\ dx_j(v) & dx_j(u) \end{vmatrix} = -\omega(v, u). \end{aligned}$$

Proof. The linearity comes from the elements of the determinant. □

Let $\Lambda^2(\mathbb{C}\mathbb{R}^n)$ denote the set of 2-forms on $\mathbb{R}^n$.

Note that $\Lambda^2(\mathbb{C}\mathbb{R}^n)$ forms a vector space over $\mathbb{R}$,

$$\Lambda^2(\mathbb{C}\mathbb{R}^n) = \{\omega : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R} \mid \omega \text{ is bilinear and skew-symmetric}\}$$

For $\omega_1, \omega_2 \in \Lambda^2(\mathbb{C}\mathbb{R}^n)$ and $\mu \in \mathbb{R}$,

$$(\omega_1 + \omega_2)(u, v) = \omega_1(u, v) + \omega_2(u, v)$$

$$\mu \cdot \omega_2(u, v) = \mu(\omega_2(u, v))$$

Exercise 10. Show the dimension of $\Lambda^2(\mathbb{C}\mathbb{R}^n)$ is given by $\binom{n}{2}$. We first show the set $\{dx_i \wedge dx_j \mid i < j\}$ forms a basis for $\Lambda^2(\mathbb{R}^n)$ (Show).

Observe $dx_i \wedge dx_i = 0$ and $dx_i \wedge dx_j = -dx_j \wedge dx_i$ and $\{dx_i\}_{i=1}^n$ is the basis of $T^*\mathbb{R}^n$.

Definition 0.52. An **exterior k -form** on $\mathbb{R}^n$ is a k -linear map

$$\omega: \underbrace{\mathbb{R}^n \times \mathbb{R}^n \times \cdots \times \mathbb{R}^n}_{k\text{-fold product}} \rightarrow \mathbb{R}$$

that is **skew-symmetric**, i.e.,

$$\omega(u_1, \dots, u_k) = \text{sgn}(\sigma) \omega(u_{\sigma(1)}, \dots, u_{\sigma(k)})$$

where σ is a permutation of $\{1, \dots, k\}$.

Example 0.11. On $\mathbb{R}^k$, we have an exterior k -form given by

$$\omega(u_1, \dots, u_k) = \begin{vmatrix} u_{11} & \cdots & u_{1k} \\ \vdots & \ddots & \vdots \\ u_{k1} & \cdots & u_{kk} \end{vmatrix}.$$

Note that $\omega(u_1, \dots, u_k)$ gives the k -dimensional volume of the parallelepiped formed by k -vectors $u_1, \dots, u_k$.

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Refer to chapter 7 in Arnold.

Definition 0.53. Wedge product on exterior forms.

Let ω^k and ω^l be a k -form, l -form on $\mathbb{R}^n$ respectively.

Define:

$$\omega^k \wedge \omega^l \quad (\text{an exterior } (k+l) \text{ form on } \mathbb{R}^n)$$

$$(\omega^k \wedge \omega^l)(u_1, \dots, u_k, u_{k+1}, \dots, u_{k+l})$$

$$= \sum (-1)^\nu \omega^k(u_{i_1}, \dots, u_{i_k}) \cdot \omega^l(u_{i_{k+1}}, \dots, u_{i_{k+l}})$$

where $i_1 < i_2 < \cdots < i_{k+l}$ and $\nu = \begin{cases} 1 & \text{if the permutation is odd} \\ 0 & \text{if the permutation is even} \end{cases}$

Exercise 11. Verify that the definition gives a $(k+l)$ -form.

Recall if ω_1 and ω_2 are 1-forms then, we defined $(\omega_1 \wedge \omega_2)(u_1, u_2)$ as:

$$\begin{vmatrix} \omega_1(u_1) & \omega_1(u_2) \\ \omega_2(u_1) & \omega_2(u_2) \end{vmatrix} = \omega_1(u_1)\omega_2(u_2) - \omega_1(u_2)\omega_2(u_1)$$

—

—

The wedge product $\wedge$ satisfies the following properties:

1.

$$w^k \wedge w^\ell = (-1)^{k\ell} w^\ell \wedge w^k$$

Proof. We express $w^k \wedge w^\ell$ as a summation over permutations $\sigma \in S_{k+\ell}$:

$$w^k \wedge w^\ell = \sum_{\sigma \in S_{k+\ell}} \text{sgn}(\sigma) w^k(V_{\sigma(1)} \cdots V_{\sigma(k)}) w^\ell(V_{\sigma(k+1)} \cdots V_{\sigma(k+\ell)}).$$

Swapping the indices k and ℓ introduces a sign factor $(-1)^{k\ell}$ due to the permutation of the combined set of indices. Thus,

$$w^k \wedge w^\ell = (-1)^{k\ell} w^\ell \wedge w^k.$$

□

2.

$$(w^k \wedge w^\ell) \wedge w^m = w^k \wedge (w^\ell \wedge w^m).$$

Proof. By the associativity of the wedge product, we have

$$(w^k \wedge w^\ell) \wedge w^m = \sum_{\sigma \in S_{k+\ell}} \text{sgn}(\sigma) w^k(V_{\sigma(1)} \cdots V_{\sigma(k)}) w^\ell(V_{\sigma(k+1)} \cdots V_{\sigma(k+\ell)}) \wedge w^m.$$

Expanding the wedge product further, we see that the associativity follows from the permutation properties of the wedge product itself, ensuring that the order of operations does not affect the final result. □

3.

$$(\lambda_1 w_1^k + \lambda_2 w_2^k) \wedge w^m = \lambda_1 (w_1^k \wedge w^m) + \lambda_2 (w_2^k \wedge w^m).$$

Proof. This follows directly from the linearity of the wedge product in each argument. By expanding the left-hand side, we obtain

$$(\lambda_1 w_1^k + \lambda_2 w_2^k) \wedge w^m = \lambda_1 w_1^k \wedge w^m + \lambda_2 w_2^k \wedge w^m,$$

which matches the right-hand side. □

Exercise 12. Verify the above properties hold.

—

On $\mathbb{R}^{2n}$ with coordinate $(p_1, p_2, \dots, p_n, q_1, \dots, q_n)$, we define the following 2-form

$$\omega^2 = \sum_{i=1}^n dp_i \wedge dq_i$$

where $\{dp_1, \dots, dp_n, dq_1, \dots, dq_n\}$ denote the dual basis.

1. Compute $\omega^2 \wedge \omega^2$

2. Compute $\omega^2 \wedge \omega^2 \wedge \cdots \wedge \omega^2$ k times $= (\omega^2)^k$

$$\begin{aligned}
(\omega^2)^k &= k! \sum_{1 \leq i_1 < \cdots < i_k \leq n} dp_{i_1} \wedge dq_{i_1} \wedge \cdots \wedge dp_{i_k} \wedge dq_{i_k} \\
\omega^2 \wedge \omega^2 &= \left(\sum_{i=1}^n dp_i \wedge dq_i \right) \wedge \left(\sum_{j=1}^n dp_j \wedge dq_j \right) \\
&= (dp_1 \wedge dq_1 + dp_2 \wedge dq_2 + \cdots + dp_n \wedge dq_n) \wedge (dp_1 \wedge dq_1 + dp_2 \wedge dq_2 + \cdots + dp_n \wedge dq_n) \\
&= (dp_1 \wedge dq_1) \wedge (dp_1 \wedge dq_1) + (dp_1 \wedge dq_1) \wedge (dp_2 \wedge dq_2) + \cdots + (dp_n \wedge dq_n) \wedge (dp_n \wedge dq_n) \\
&\quad + \cdots \\
&= (dp_1 \wedge dq_1) \wedge (dp_1 \wedge dq_1) + \cdots \\
&= (dp_1 \wedge dq_1 \wedge dp_1 \wedge dq_1) + \cdots \\
&= -(dp_1 \wedge dp_1 \wedge dq_1 \wedge dq_1) + \cdots \\
&= 0 + \cdots
\end{aligned}$$

Observe that,

$$(dp_i \wedge dq_i) \wedge (dp_i \wedge dq_i) = 0$$

for all $1 \leq i \leq n$.

But, for $i \neq j$,

$$(dp_i \wedge dq_i) \wedge (dp_j \wedge dq_j)$$

appears twice.

$$\therefore \omega^2 \wedge \omega^2 = 2 \sum_{i < j} dp_i \wedge dq_i \wedge dp_j \wedge dq_j$$

Given a vector $u \in \mathbb{R}^3$, we consider the 2-form

$$\omega_u(v_1, v_2) = u \cdot (v_1 \times v_2) \quad \text{for } v_1, v_2 \in \mathbb{R}^3.$$

To find the wedge product of ω_{u_1} and ω_{u_2} , we compute:

$$\omega_{u_1} \wedge \omega_{u_2}(v_1, v_2, v_3, v_4) = \sum_{\sigma \in S_4} \text{sgn}(\sigma) \omega_{u_1}(v_{\sigma(1)}, v_{\sigma(2)}) \omega_{u_2}(v_{\sigma(3)}, v_{\sigma(4)})$$

Expanding this, we have:

$$\sum_{\sigma \in S_4} \text{sgn}(\sigma) (u_1 \cdot (v_{\sigma(1)} \times v_{\sigma(2)})) (u_2 \cdot (v_{\sigma(3)} \times v_{\sigma(4)}))$$

There is no non-trivial 4-form for dimensions in $\mathbb{R}^3$, so we get zero.

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and ω^k be an exterior k -form on $\mathbb{R}^m$.

We define a new exterior k -form on $\mathbb{R}^n$ as follows:

Definition 0.54. We denote it by $T^*\omega^k$.

$$T^*\omega^k(v_1, \dots, v_k) = \omega^k(Tv_1, \dots, Tv_k)$$

is called the **pullback form** via T or **pullback** of ω^k via T .

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function. For a point $x \in \mathbb{R}^n$, the derivative of f ,

$$df_x : T_x\mathbb{R}^n \rightarrow \mathbb{R}$$

is a linear map. The derivative is a smooth map $df : T\mathbb{R}^n \rightarrow \mathbb{R}$ such that for each point we get an ext. 1-form

$$df_x : T_x\mathbb{R}^n \rightarrow \mathbb{R}.$$

Definition 0.55. In general we define a differential 1-form as follows. Let M be a manifold of dimension n . Let $TM = \bigsqcup_{x \in M} T_xM$ denote the tangent bundle of M . A differential 1-form on M is a smooth map

$$\omega : TM \rightarrow \mathbb{R}$$

such that for each $x \in M$, we have

$$\omega|_{T_xM} = \omega_x : T_xM \rightarrow \mathbb{R}$$

an ext. 1-form.

Example 0.12. Let $f : M \rightarrow \mathbb{R}$ be a smooth function, then $df : TM \rightarrow \mathbb{R}$ is a differential 1-form.

Theorem 0.2. Let $(x_1, \dots, x_n)$ denote a coordinate system on $\mathbb{R}^n$. Let ω be a differential 1-form, then

$$\omega = \sum_{i=1}^n q_i(x_1, \dots, x_n) dx_i$$

where q_i 's are smooth real-valued functions on $\mathbb{R}^n$.

Proof. For each $x = (x_1, \dots, x_n)$, the differential form ω restricts to be an exterior 1-form on $T_x\mathbb{R}^n$. That is,

$$\omega_x : T_x\mathbb{R}^n \rightarrow \mathbb{R}.$$

We know that $\omega_x = \sum_{i=1}^n c_i dx_i$, where c_i are scalars depending on x . The smoothness of ω implies that the maps

$$x \mapsto c_i(x)$$

are smooth. □

Corollary 0.1. Let M be a smooth manifold of dimension n and (U, φ) be a chart. Suppose $(x_1, \dots, x_n)$ denote the coordinate system given by (U, φ) . Then a differential 1-form ω can be equipped as follows in the coordinate chart (U, φ) :

$$\omega = \sum_{i=1}^n a_i(x_1, \dots, x_n) dx_i,$$

where a_i 's are smooth functions defined on the chart (U, φ) .

Definition 0.56. A differential 2-form on $\mathbb{R}^n$ is a smooth map

$$\omega^2$$

such that for each $x \in \mathbb{R}^n$, ω_x^2 is an exterior 2-form on $T_x \mathbb{R}^n$.

Theorem 0.3. If ω^2 is a differential 2-form on $\mathbb{R}^n$, then

$$\omega^2 = \sum_{i < j} q_{ij}(x) dx_i \wedge dx_j \quad (*)$$

where $q_{ij} : \mathbb{R}^n \rightarrow \mathbb{R}$ are smooth functions.

Example 0.13. On $\mathbb{R}^2$ with (x, y) coordinates, we have the area form given by $dx \wedge dy$.

Exercise 13. Change the coordinates to polar coordinates described by (r, Θ) where $x^2 + y^2 = r^2$ and $\tan \Theta = \frac{y}{x}$. How does the area form look like in (r, Θ) coordinates?

We have the following relationships:

$$x^2 + y^2 = r^2, \quad \tan \Theta = \frac{y}{x}$$

Consider the differentials:

$$2r dr = 2x dx + 2y dy = 2(x dx + y dy)$$

For the angle Θ , we have:

$$\sec^2 \Theta d\Theta = \frac{x dy - y dx}{x^2}$$

Using the identity $1 + \tan^2 \Theta = \sec^2 \Theta$, we get:

$$(1 + \tan^2 \Theta) d\Theta = \frac{x dy - y dx}{x^2}$$

Since $x^2 + y^2 = r^2$, we can rewrite the above as:

$$\frac{x^2 + y^2}{x^2} d\Theta = \frac{x dy - y dx}{x^2}$$

This simplifies to:

$$d\Theta = \frac{x dy - y dx}{x^2 + y^2}$$

We also have:

$$r dr = x dx + y dy$$

Multiplying these results, we get:

$$r^2 d\Theta = x dy - y dx$$

Now, consider the wedge product of $r dr$ and $d\Theta$:

$$r^3 dr \wedge d\Theta = (x dx + y dy) \wedge (x dy - y dx)$$

Expanding the wedge product:

$$r^3 dr \wedge d\Theta = x^2 dx \wedge dy - y^2 dy \wedge dx$$

Since $dx \wedge dy = -dy \wedge dx$, we have:

$$r^3 dr \wedge d\Theta = x^2 dx \wedge dy + y^2 dx \wedge dy$$

This simplifies to:

$$r^3 dr \wedge d\Theta = (x^2 + y^2) dx \wedge dy$$

Given that $x^2 + y^2 = r^2$, we finally obtain:

$$r dr \wedge d\Theta = dx \wedge dy$$

Definition 0.57. A differential 2-form ω^2 on M is a smooth map that restricts to each $T_x M$, for each x , to give an exterior 2-form. We denote it by ω_x .

In a chart $(U, \varphi = (x_1, \dots, x_n))$ on M , ω^2 is expressed in $(*)$.

On $\mathbb{R}^n$ with coordinates $(x_1, \dots, x_n)$, a basis of exterior k -forms is given by

$$\{dx_{i_1} \wedge \dots \wedge dx_{i_k} \mid i_1, \dots, i_k \in \{1, \dots, n\} \text{ s.t. } i_1 < i_2 < \dots < i_k\}.$$

For $dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_k}(u_1, u_2, \dots, u_k)$, we have

$$dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_k}(u_1, u_2, \dots, u_k) = \sum_{\sigma} \text{sgn}(\sigma) dx_{i_1}(u_{\sigma(1)}) dx_{i_2}(u_{\sigma(2)}) \dots dx_{i_k}(u_{\sigma(k)}).$$

Suppose (U, φ) is a chart for M . Let $(x_1, \dots, x_n)$ denote the coordinate system from (U, φ) .

Then,

$$\omega^k = \sum_{1 \leq i_1 \leq i_2 \leq \dots \leq i_k \leq n} a_{i_1 i_2 \dots i_k}(x_1, \dots, x_n) dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_k},$$

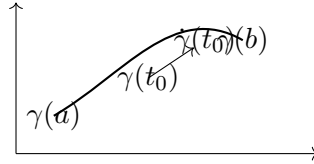
where each $a_{i_1 i_2 \dots i_k}$ is a smooth real-valued function.

Integration of forms.

Let ω be a differential 1-form. Given a smooth curve $\gamma : [a, b] \rightarrow M$, the curve γ is a smooth curve $\tilde{\gamma} : (a - \varepsilon, b + \varepsilon) \rightarrow M$. For each point x on γ , we have the number $\omega_x(\dot{\gamma}(t))$ where $\dot{\gamma}(t)$ is the tangent vector at x .

The integral of ω along γ is defined as:

$$\int_{\gamma} \omega = \int_a^b \omega_{\gamma(t)}(\dot{\gamma}(t)) dt$$



Integral of n -forms

Let ω^n denote a differential n -form on $\mathbb{R}^n$.

Definition 0.58. A differential n -form ω^n on $\mathbb{R}^n$ can be expressed as:

$$\omega^n = f(x_1, \dots, x_n) dx_1 \wedge dx_2 \wedge \dots \wedge dx_n,$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a smooth function.

For a point $p \in \mathbb{R}^n$,

$$\omega_p^n = f(p) dx_1 \wedge dx_2 \wedge \dots \wedge dx_n.$$

The integral of ω^n over a polyhedron Δ^n is given by:

$$\int_{\Delta^n} \omega^n = \int_{\Delta^n} f(x_1, \dots, x_n) dx_1 \wedge dx_2 \wedge \dots \wedge dx_n,$$

where Δ^n is a polyhedron which is a compact and convex n -dimensional subset of $\mathbb{R}^n$.

Definition 0.59. Let M be an n -dimensional manifold. We say M is *oriented* if we can consistently orient n -dimensional parallelepipeds on M .

Let ω^n denote a differential n -form on M .

Theorem 0.4. Assume M is compact. Then M can be broken into a finite collection of nicely fitting n -dimensional polyhedra, denoted by $S = \{\Delta_1^n, \Delta_2^n, \dots, \Delta_k^n\}$. The integral of ω^n over M is given by the sum of the integrals of ω^n over each polyhedron in S :

$$\int_M \omega^n = \sum_{\Delta^n \in S} \int_{\Delta^n} \omega^n.$$

Let ω^k denote a differential k -form on a smooth manifold M of dimension n .

Definition 0.60. Let Δ^k denote a convex polyhedron in $\mathbb{R}^k$. Specifically, Δ^k is the convex hull of $(k+1)$ points in $\mathbb{R}^k$.

Definition 0.61. A k -cell σ in M is a triple (Δ^k, f, Or) , where:

- Δ^k is a convex polyhedron in $\mathbb{R}^k$,
- $f : \Delta^k \rightarrow M$ is a smooth function,
- Or denotes an orientation on Δ^k .

Example 0.14. An (oriented) path is a 1-cell, given by a smooth function $f : [a, b] \rightarrow M$.

Definition 0.62. A k -chain is a formal integral combination of k -forms. If $\sigma_1, \sigma_2, \dots, \sigma_r$ are k -cells, then a k -chain is a formal integral combination

$$\sum_{i=1}^r m_i \sigma_i,$$

where $m_i \in \mathbb{Z}$.

Remark 0.19. Note: Each t_i is a 2-cell in the annulus.

Then, $T_1 + T_2 + \dots + T_6$ is a 2-chain in the annulus.

Another 2-chain is $2T_1 + 4T_2 + 6T_3 + 7T_4$.

$$C^k = \sum_{i=1}^r m_i \sigma_i \text{ is defined as}$$

$$\int_{C^k} \omega^k = \sum_{i=1}^r m_i \int_{\sigma_i} \omega^k$$

Exterior derivative of k -forms

Let M be a smooth manifold of dimension n .

Motivation:

Consider a smooth function $\varphi : [a, b] \rightarrow \mathbb{R}$. We consider φ as a 0-form.

In this example, 0-chains are formal finite combinations of points.

In particular, we have formal combinations of the type

$$\sum_{i=1}^r n_i \cdot x_i, \quad \text{where } x_i \text{ are points in the space } [a, b].$$

The integral of a 0-form φ on the 0-chain

$$\sum_{i=1}^r n_i \cdot x_i$$

is given by

$$\sum_{i=1}^r n_i \varphi(x_i).$$

In particular, for an interval $[x, y] \subseteq [a, b]$, consider the chain $y - x$.

The integral of 0-form φ on the chain $y - x$ is given

$$\int \varphi' = \varphi(y) - \varphi(x).$$

If we take $F = \varphi'$, then φ' can be treated as the “derivative” of 0-form φ . In particular, φ' is a 1-form in the sense of integral of 1-form on $[x, y]$.

Suppose ω^k is a k -form on M , we want

$$\int_{\sigma} \omega^k = \int_{\partial c} \Theta^{k+1} \quad \text{to hold,}$$

where σ is a k -chain that forms the boundary of the k -cell.

Example 0.15. Let V be a vector field on $\mathbb{R}^3$. The flux of V passing through S is given by

$$\int_S \vec{V} \cdot d\vec{n} = \int_S \text{div } V \, dx \, dy \, dz.$$

The above example suggests that we take the derivative of $\int \omega^k$ to get Ω^{k+1} .

Let $\sigma = \omega c$.

Definition 0.63. Let $(u_1, u_2, \dots, u_{k+1})$ denote a $(k+1)$ -parallelepiped in $\mathbb{R}^n$. In particular, it forms $(k+1)$ -cells. Take the integral ω^k on the boundary of this $(k+1)$ -cell. We consider the variation coming from each edge of $(u_1, \dots, u_k)$.

Then consider the limit

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon^{k+1}} \int_{\partial \varepsilon u_1, \dots, \varepsilon u_{k+1}} \omega^k.$$

Definition 0.64. If ω^k is a k -form on $\mathbb{R}^n$ with coordinate system $(x_1, x_2, \dots, x_n)$, then

$$\omega^k = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} a_{i_1 i_2 \dots i_k}(x_1, \dots, x_n) dx_{i_1} \wedge \dots \wedge dx_{i_k}$$

where $a_{i_1 i_2 \dots i_k}$ is a smooth function on $\mathbb{R}^n$.

Definition 0.65. Define the exterior derivative of ω^k to be the following:

$$d\omega^k = \sum_{1 \leq i_1 < \dots < i_k \leq n} da_{i_1 i_2 \dots i_k} \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k}$$

where,

$$da_{i_1 i_2 \dots i_k} = \left(\sum_{j=1}^n \frac{\partial}{\partial x_j} a_{i_1 i_2 \dots i_k}(x_1, \dots, x_n) dx_j \right).$$

Example 0.16. In practice, the definition works as follows:

$$\begin{aligned}\omega^1 &= xy \, dz + xz \, dy + yz \, dx \\ d\omega^1 &= d(xy \, dz) + d(xz \, dy) + d(yz \, dx) \\ &= y \, dx \wedge dz + x \, dy \wedge dz + z \, dx \wedge dy + x \, dz \wedge dy + y \, dz \wedge dx + z \, dy \wedge dx \\ &= 0\end{aligned}$$

Let X be a smooth manifold of dimension n .

Definition 0.66. $\Omega^k(X) :=$ The vector space of differential k -forms.

Definition 0.67. $\Omega^0(X) :=$ The space of smooth $\mathbb{R}$ -valued functions on X .

We have the exterior differentiation on k -forms

$$d : \Omega^k(X) \rightarrow \Omega^{k+1}(X)$$

is a $\mathbb{R}$ -linear map.

1. Take $X = \mathbb{R}^n$ with the coordinate system $(x_1, \dots, x_n)$. Consider the 1-form given by

$$\alpha = \alpha_1 \, dx_2 + \alpha_3 \, dx_4.$$

Find $d\alpha$.

Proof.

$$\begin{aligned}d\alpha &= d(\alpha_1 \, dx_2 + \alpha_3 \, dx_4) \\ &= \frac{\partial \alpha_1}{\partial x_1} \, dx_1 \wedge dx_2 + \frac{\partial \alpha_3}{\partial x_3} \, dx_3 \wedge dx_4.\end{aligned}$$

□

—

and the rest are $\frac{\partial f_i}{\partial x_i} = 0$.

Hence, we get

$$dx_1 \wedge dx_2 + dx_3 \wedge dx_4$$

- (2) Take $X = \mathbb{R}^3$ with (x, y, z) coordinates.

Let $\beta = dz - y \, dx$.

Compute $\beta \wedge d\beta$.

$$d\beta = d(dz - y \, dx) = 0 - \left(\frac{\partial y}{\partial x} \, dx \wedge dx + dy \wedge dx \right) = dx \wedge dy$$

Now $\beta \wedge d\beta = (dz - y \, dx) \wedge (dx \wedge dy)$

$$= dz \wedge dx \wedge dy - y \, dx \wedge dx \wedge dy = dx \wedge dy \wedge dz.$$

—

Theorem 0.5. The composition of successive exterior derivatives is zero. In particular,

$$\Omega^k(X) \xrightarrow{d} \Omega^{k+1}(X) \xrightarrow{d} \Omega^{k+2}(X),$$

the map $d^2 \equiv 0$.

Proof. Verify the above statement for $d\alpha$ and β . That is, show

$$d(d\alpha) = d(d\alpha_1 \wedge d\alpha_2 + d\alpha_3 \wedge d\alpha_4) + d(d\alpha_5 \wedge d\alpha_6) + d(d\alpha_7 \wedge d\alpha_8).$$

(note the exterior derivative is linear).

For $d(d\alpha_1 \wedge d\alpha_2)$:

$$d(d\alpha_1 \wedge d\alpha_2) = d(1 \wedge d\alpha_1 \wedge d\alpha_2) = \left(\sum_{j=1}^n \frac{\partial}{\partial x_j} d\alpha_j \right) \wedge d\alpha_1 \wedge d\alpha_2 = 0.$$

Similarly,

$$d(d\alpha_3 \wedge d\alpha_4) = 0.$$

□

$$d(dB) = d(da \wedge dy \wedge dz)$$

$$= d(1) \wedge da \wedge dy \wedge dz$$

$$= \left(\frac{\partial(1)}{\partial x} dx + \frac{\partial(1)}{\partial y} dy + \frac{\partial(1)}{\partial z} dz \right) \wedge da \wedge dy \wedge dz$$

$$= 0.$$

In particular, if $f : X \rightarrow \mathbb{R}$ is a smooth function, then $d(df) = 0$.

It is sufficient to verify that $d^2 f = 0$ in any given chart on X .

—

Thus, we may assume that $f : V \rightarrow \mathbb{R}$ is a function with given input given by $(x_1, \dots, x_n)$. Therefore,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

Now apply d again on the above equation:

$$d(df) = d\left(\sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i\right).$$

Using the linearity of d , we have

$$d(df) = \sum_{i=1}^n d\left(\frac{\partial f}{\partial x_i}\right) \wedge dx_i.$$

For each i ,

$$d\left(\frac{\partial f}{\partial x_i}\right) = \sum_{j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j} dx_j.$$

Substituting back, we get

$$\begin{aligned} d(df) &= \sum_{i=1}^n \left(\sum_{j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j} dx_j \right) \wedge dx_i \\ &= \sum_{i \neq j} \left(\frac{\partial^2 f}{\partial x_i \partial x_j} dx_j \wedge dx_i \right) \\ &= \sum_{i \neq j} \left(\frac{\partial^2 f}{\partial x_i \partial x_j} - \frac{\partial^2 f}{\partial x_j \partial x_i} \right) dx_i \wedge dx_j. \end{aligned}$$

—

Note: We are generalizing that curl of divergence of (smooth) potential is zero.

Proof. Take k -form α and express in chart (U, φ) with coordinate system $(\alpha_1, \dots, \alpha_n)$:

$$\alpha = \sum_{1 \leq i_1 < \dots < i_k \leq n} f_{i_1, \dots, i_k}(\alpha_1, \dots, \alpha_n) d\alpha_{i_1} \wedge \dots \wedge d\alpha_{i_k}.$$

Then

$$\begin{aligned} d\alpha &= \sum_{1 \leq i_1 < \dots < i_k \leq n} \left(\sum_{j=1}^n \frac{\partial f_{i_1, \dots, i_k}(\alpha_1, \dots, \alpha_n)}{\partial \alpha_j} \right) d\alpha_j \wedge d\alpha_{i_1} \wedge \dots \wedge d\alpha_{i_k} \\ &= \sum_{1 \leq i_1 < \dots < i_k \leq n} \sum_{j \notin \{i_1, \dots, i_k\}} \frac{\partial f_{i_1, \dots, i_k}}{\partial \alpha_j} d\alpha_j \wedge d\alpha_{i_1} \wedge \dots \wedge d\alpha_{i_k} \end{aligned}$$

$$\begin{aligned}
&= \sum_{j \notin \{i_1, \dots, i_k\}} \sum_{1 \leq i_1 < \dots < i_k \leq n} \frac{\partial f_{i_1, \dots, i_k}}{\partial \alpha_j} d\alpha_j \wedge d\alpha_{i_1} \wedge \dots \wedge d\alpha_{i_k} \\
&\implies d(d\alpha) = \sum \left(d \left(\frac{\partial f_{i_1, \dots, i_k}}{\partial \alpha_j} \right) \wedge d\alpha_j \wedge d\alpha_{i_1} \wedge \dots \wedge d\alpha_{i_k} \right) \\
&= \sum \left(\sum_{\ell=1}^n \frac{\partial^2 f_{i_1, \dots, i_k}}{\partial \alpha_j \partial \alpha_\ell} d\alpha_\ell \right) \wedge d\alpha_j \wedge d\alpha_{i_1} \wedge \dots \wedge d\alpha_{i_k}.
\end{aligned}$$

□

$$\sum_{1 \leq i_1 < \dots < i_k \leq n} \left(\frac{\partial^2 f_{i_1, \dots, i_k}}{\partial x_j \partial x_\ell} - \frac{\partial^2 f_{j, i_2, \dots, i_k}}{\partial x_i \partial x_\ell} \right) dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_j = 0$$

Definition 0.68. A k -form is closed if $d\alpha = 0$.

Example 0.17. Any k -form that is an exterior derivative of a $(k-1)$ form is closed.

If $\beta \in \Omega^k(X)$ such that $\beta = d\alpha$ for some $\alpha \in \Omega^{k-1}(X)$, then

$$d\beta = d(d\alpha) = 0.$$

On $\mathbb{R}^n$, the converse is also true. In particular, if β is a closed k -form ($d\beta = 0$), then there is a $(k-1)$ form α such that

$$\beta = d\alpha.$$

Definition 0.69. Let X be a smooth manifold. If ω is a (smooth) 2-form on X such that

- (1) $d\omega = 0$ (ω is closed),
- (2) ω is non-degenerate when restricted to each tangent space $T_x X$,

then, we call ω a *symplectic form* on X . The pair (X, ω) is called a *symplectic manifold*.

Example 0.18. Consider the symplectic manifold $(\mathbb{R}^2, dx \wedge dy)$:

- 1. $d(dx \wedge dy) = 0$.

Proof. We verify that $dx \wedge dy$ when restricted to a tangent space $T_p \mathbb{R}^2 \cong \mathbb{R}^2$ is non-degenerate.

$$dx \wedge dy|_{T_p \mathbb{R}^2} : T_p \mathbb{R}^2 \times T_p \mathbb{R}^2 \rightarrow \mathbb{R}$$

For vectors $u = (u_1, u_2)$ and $v = (v_1, v_2)$,

$$dx \wedge dy(u, v) = \begin{vmatrix} u_1 & v_1 \\ u_2 & v_2 \end{vmatrix}.$$

□

Definition 0.70. A bilinear form ω on a vector space V is non-degenerate if for $u \neq 0$, a vector in V , there exist $v \in V$ such that $\omega(u, v) \neq 0$.

Example 0.19. An example of a degenerate bilinear form is given by

$$\beta: \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$$

defined as

$$\beta(u, v) = \begin{vmatrix} u_1 & v_1 \\ u_2 & v_2 \end{vmatrix}.$$

Let β be a bilinear form on V . ($\beta: V \times V \rightarrow \mathbb{R}$ is a bilinear map.)

Fix a vector $u \in V$. Then consider the functional

$$I(u): V \rightarrow \mathbb{R}$$

as follows:

$$I(u)(v) = \beta(u, v).$$

We have

$$I(u_1 + u_2) = I(u_1) + I(u_2).$$

Define a linear map as follows:

$$\begin{array}{ccc} I: & V & \rightarrow \\ & V^* & \\ & u & \mapsto \\ & I(u). & \end{array}$$

Exercise 14. Let I be an isomorphism of vector spaces if and only if B is a non-degenerate form on V .

Given:

$$I: V \rightarrow V^*, \quad u \mapsto I(u)(\cdot) = B(Cu, \cdot)$$

Proof. ($\Rightarrow$) WTS: for any $(u \neq 0) \in V$, there exists $v \in V$ such that $B(u, v) \neq 0$.

Assume there exists a nonzero $u \in V$ such that $I(u)(v) = 0$ for all $v \in V$; then $u \in \ker I$.

Observe that $\ker I = \{0\}$ as I is an isomorphism.

($\Leftarrow$) It is sufficient to show I is an injective linear map. As linear maps in finite dimension are injective if and only if they are surjective.

Linearity can be determined from the bilinear form B , whereas if injectivity fails, then B is a degenerate form. Hence, I is an isomorphism. $\square$

The map I is induced by symplectic form is an isomorphism between $T_p X$ and $T_p^* X$ for each point $p \in X$.

In general, the rank (I) is referred to as the rank of the bilinear form $\rho = \dim$ of range of I .

Example 0.20. Coming back to example (1), verify $dx \wedge dy$ is a non-degenerate form.

For any $u \neq 0$, consider v obtained by rotating u by 90° in the anti-clockwise direction.

Claim 7. $dx \wedge dy(u, v) = \|u\|^2$

Proof. Let

$$u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \quad v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$\text{If } v = \begin{pmatrix} -u_2 \\ u_1 \end{pmatrix},$$

□

—

Consider the 2-form on $\mathbb{R}^{2n}$ given by

$$dx \wedge dy(u, v) = \begin{vmatrix} u_1 & -u_2 \\ u_2 & u_1 \end{vmatrix} = u_1^2 + u_2^2.$$

1. On $(\mathbb{R}^{2n}, (p_i, q_i) \mid 1 \leq i \leq n)$ consider the 2-form given by

$$\omega = \sum_{i=1}^n dp_i \wedge dq_i.$$

In order to show that the above is an example of a bilinear form, we prove the following:

(a) *Proof.*

$$d\omega = d\left(\sum_{i=1}^n dp_i \wedge dq_i\right) = \sum_{i=1}^n d(dp_i \wedge dq_i) = 0.$$

□

(b) ω is non-degenerate.

Proof. Consider the map

$$\sum_{i=1}^n dp_i \wedge dq_i : T_x \mathbb{R}^{2n} \times T_x \mathbb{R}^{2n} \rightarrow \mathbb{R}.$$

In particular, for $\alpha = (p_1, q_1, \dots, p_n, q_n)$, choose $\beta = (-q_1, p_1, -q_2, p_2, \dots, -q_n, p_n)$. Each 2×2 determinant will be positive. □

—

Theorem 0.6. The following are equivalent:

1. The rank of B is n ($= \dim V$).
2. The bilinear form B is non-degenerate.
3. The linear map $V \rightarrow V^*$ is an isomorphism.

Example 0.21. The dot product of vectors in $\mathbb{R}^n$ is a non-degenerate bilinear form.

$$\begin{aligned}\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n &\rightarrow \mathbb{R} \\ (u, v) &\mapsto \langle u, v \rangle.\end{aligned}$$

The corresponding induced map is given by

$$\begin{aligned}I(u) : \mathbb{R}^n &\rightarrow \mathbb{R} \\ v &\mapsto \langle u, v \rangle.\end{aligned}$$

Suppose $\{e_1, \dots, e_n\}$ is the standard basis of $\mathbb{R}^n$. Thus $I(e_i) = e_i^*$,

$$e_i^*(v) = \langle v, e_i \rangle = v_i.$$

For $(\mathbb{R}^{2n}, \omega)$, we have an induced isomorphism

$$I : \mathbb{R}^{2n} \rightarrow (\mathbb{R}^{2n})^*$$

Suppose $\mathbb{R}^{2n}$ is given by a basis

$$\begin{aligned}&\{p_1, q_1, \dots\} \\ p_1 &= (1, 0, \dots, 0) \\ p_2 &= (0, 0, 1, \dots, 0) \\ p_i &= (0, \dots, 1, \dots, 0) \quad \text{at the } (2i-1)^{\text{th}} \text{ position} \\ q_1 &= (0, 1, \dots, 0) \\ q_2 &= (0, 0, 0, 1, \dots, 0) \\ q_i &= (0, \dots, 1, \dots, 0) \quad \text{at the } (2i)^{\text{th}} \text{ position}\end{aligned}$$

$$\omega = \sum_{i=1}^n dp_i \wedge dq_i$$

Exercise 15. Find $I(p_i)$.

Proof. Figure out algebraically.

$$\begin{aligned}I(p_i)(p_j) &= \omega(p_i, p_j) = 0 \\ I(p_i)(q_j) &= \omega(p_i, q_j) = \delta_{ij}\end{aligned}$$

□

We define the map $p_i^* = I \circ p_i$.

Under identification by I , p_i^* is identified with q_i and observe, $q_i^* = -p_i$.

Exercise 16. On $\mathbb{R}^3$, does there exist a symplectic bilinear form? (skew-symmetric, non-degenerate bilinear form).

Assume B to be a skew-symmetric, non-degenerate bilinear form.

Observe let e_1, e_2, e_3 be basis elements of $\mathbb{R}^3$. We construct matrix A

$$A = (B(e_i, e_j)) \text{ for } 1 \leq i, j \leq 3.$$

Observe $B(u, v) = u^T A v$. Now if B is non-degenerate then A is invertible.

Fix a v such that

$$u^T A v = 0 \implies u = 0 \text{ then } v = 0 \text{ as } B \text{ is non-degenerate.}$$

Now, we have $Av = 0 \implies u^T Av = 0 \implies u^T uv = 0$. Then $v = 0$ and $v = 0 \implies Av = 0$.

Thus, $Av = 0$ if and only if $v = 0 \implies A$ is invertible.

Proof. By contradiction: $\det A = 0 \implies A$ is non-invertible. Hence, no such form exists. $\square$

Definition 0.71. Let (V, ω) be a symplectic linear space. Let W be a subspace of V . We call W to be **isotropic** if

$$W \subseteq W^\perp = \{v \in V \mid \omega(v, w) = 0 \ \forall w \in W\}.$$

Example 0.22. Any 1-dimensional subspace of V is isotropic.

Proof. Assume that $W = \langle w_0 \rangle$, then

$$W^\perp := \{v \in V \mid \omega(v, w_0) = 0\}.$$

Since W is 1-dimensional, for any $w \in W$, $w = \lambda w_0$ for some $\lambda \in \mathbb{R}$. Then,

$$\omega(w, w) = \omega(\lambda w_0, \lambda w_0) = \lambda^2 \omega(w_0, w_0) = 0,$$

because ω is skew-symmetric and $\omega(w_0, w_0) = 0$. Thus, $W \subseteq W^\perp$. $\square$

Remark 0.20. Properties of $W^\perp$:

- $W^\perp = \{v \in V \mid \omega(v, w) = 0 \ \forall w \in W\}$.
- $W^\perp$ is a subspace of V .
- Observe that $\dim W + \dim W^\perp = \dim V$.

Proof of the observation. Consider the map

$$I: V \rightarrow W^*,$$

$$v \mapsto \omega(v, \cdot): W \rightarrow \mathbb{R}.$$

The kernel of I is $W^\perp$. By the rank-nullity theorem,

$$\dim V = \dim \ker I + \dim \operatorname{Im} I = \dim W^\perp + \dim \operatorname{Im} I.$$

Since $\operatorname{Im} I \cong W^*$ and $\dim W^* = \dim W$, we have

$$\dim V = \dim W^\perp + \dim W.$$

□

Given the content in the image, —

Let W^* be the dual space of W and let $I : V \rightarrow W^*$ be an isomorphism.

Definition 0.72. Let $B : W \rightarrow \mathbb{R}$ be a functional.

Proof. Goal: To find $v_0 \in V$ such that $B(w) = \omega(v_0, w)$ for all $w \in W$.

Extend B on V , all this extension $\tilde{B}$ is defined as follows:

$$\tilde{B} : V \rightarrow \mathbb{R}$$

such that

$$\tilde{B}(w) = B(w) \quad \forall w \in W.$$

By non-degeneracy (by the fact $I : V \rightarrow V^*$ is an isomorphism), there exists $v_0 \in V$ such that

$$\tilde{B}(u) = \omega(v_0, u).$$

By the Rank-Nullity theorem,

$$\dim W + \dim W^\perp = \dim V.$$

□

—

For any isotropic subspace W of (V, ω) :

$$(W^\perp)^\perp = W \quad (\text{This is true for every subspace})$$

$$W \subseteq W^\perp \implies (W^\perp)^\perp \subseteq W^\perp$$

$$[V \subseteq W \implies W^\perp \subseteq V^\perp, \text{ for general subspace}]$$

$$\dim W^\perp = n - \dim W$$

$$\dim(W^\perp)^\perp = n - \dim W^\perp$$

$$= n - (n - \dim W)$$

$$= \dim W$$

Proof. Verify that $W \subseteq (W^\perp)^\perp$.

Observe that $\dim(W^\perp)^\perp = \dim W$.

Hence, $(W^\perp)^\perp = W$. □

Exercise 17. If W is isotropic, then $\dim W \leq \frac{1}{2}(\dim V)$.

Proof. Since $W \subseteq W^\perp$, we have $\dim W \leq \dim W^\perp$.

Now, $\dim W^\perp + \dim W = \dim V$, this implies

$$2 \dim W \leq \dim V \implies \dim W \leq \frac{1}{2}(\dim V).$$

□

Exercise ($\mathbb{R}^4, \omega = dp_1 \wedge dq_1 + dp_2 \wedge dq_2$) Find the general form of 2-dimensional isotropic subspaces.

Let u, v be a basis of W i.e. $\dim 2$ space

$$\begin{vmatrix} u_{p_1} & v_{p_1} \\ u_{p_2} & v_{p_2} \end{vmatrix} + \begin{vmatrix} u_{p_3} & v_{p_3} \\ u_{p_4} & v_{p_4} \end{vmatrix} = 0$$

$$= u_{p_1}v_{p_2} - v_{p_1}u_{p_2} + u_{p_3}v_{p_4} - v_{p_3}u_{p_4} = 0$$

$$u_{p_1}v_{p_2} + u_{p_3}v_{p_4} = v_{p_1}u_{p_2} + v_{p_3}u_{p_4}$$

Definition 0.73. A subspace $W \subset (V, \omega)$ is called a **Lagrangian** if W is a maximal isotropic subspace (or equivalently W is isotropic and $\dim W = \dim V/2$).

Natural Symplectic Structures on Cotangent Bundle

Take $X = \mathbb{R}^n$ with coordinate $(x_1, \dots, x_n)$. Then,

$$T^*X = X \times T_x^*X = X \times (\mathbb{R}^n)^*, 0$$

where 0 denotes the origin in $\mathbb{R}^n$ with coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$.

We describe a natural 1-form on T^*X :

$$\lambda_{(x, \alpha)}: T_{(x, \alpha)}(T^*X) \rightarrow \mathbb{R}$$

Observation

$$T_{(x,\alpha)}(T^*X) = T_{(x,\alpha)}(X \times T_x^*X) = T_x(X) \times T_\alpha(T_x^*X)$$

$$[T_x^*X \text{ is a vector space is identified with } T_\alpha(T_x^*X)]$$

$$\begin{aligned} \lambda_{(x,\alpha)}: T_x X \times T_x^* X &\rightarrow \mathbb{R} \\ (u, \beta) &\mapsto \alpha(u) \end{aligned}$$

Exercise:

Show that $\lambda_{(x,\alpha)}: T_{(x,\alpha)}(T^*X) \rightarrow \mathbb{R}$ is a linear map.

Note that the map $(x, \alpha) \mapsto \lambda_{(x,\alpha)}$ is smooth as the dependence on the x component is smooth and the dependence on the α component is linear and hence smooth.

Consider the 2-form obtained by taking the exterior derivative of λ . The desired 2-form is $d\lambda$.

Exercise 18. Work out the expression for λ and $d\lambda$ when $X = \mathbb{R}^n$.

Let $T^*X = \coprod_{(x_1, \dots, x_n, y_1, \dots, y_n) \in \mathbb{R}^{2n}} \{(\alpha_1, \dots, \alpha_n, y_1, \dots, y_n)\}$ for all $1 \leq i \leq n$.

Define the map

$$T_{(x_1, \dots, x_n, y_1, \dots, y_n)}: T_{(x_1, \dots, x_n)}(T^*X) \rightarrow \mathbb{R}$$

by

$$T_{(x_1, \dots, x_n, y_1, \dots, y_n)}(u_1, \dots, u_n, \beta_1, \dots, \beta_n) = y_1 u_1 + \dots + y_n u_n.$$

Then,

$$\lambda_{(x_1, \dots, x_n, y_1, \dots, y_n)} = y_1 d\alpha_1 + \dots + y_n d\alpha_n.$$

The 1-form obtained is called the canonical 1-form α on T^*X . The 2-form is given by

$$d\lambda = dy_1 \wedge d\alpha_1 + \dots + dy_n \wedge d\alpha_n.$$

This 2-form is called the canonical symplectic form.

Theorem 0.7. Let X be a smooth manifold of dimension n . There exists a canonical / natural symplectic form on T^*X .

Proof. Consider the projection map

$$\pi: T^*X \rightarrow X, \quad (\underline{x}, \underline{a}) \mapsto \underline{x}.$$

Then, $d\pi: T(T^*X) \rightarrow TX$.

$$[\text{Recall } d\pi_{(\underline{x}, \underline{a})}: T_{(\underline{x}, \underline{a})}(T^*X) \rightarrow T_{\underline{x}}X]$$

Define 1-form as follows,

$$\lambda_{(\underline{x}, \underline{a})}(v) = \underline{a}(\pi_{(\underline{x}, \underline{a})}(v)).$$

Exercise 19. Verify that $\lambda_{(\underline{x}, \underline{a})}$ is linear in the variable v . $\underline{a}$ is linear.

Exercise 20. Show that the above definition applied on $X = \mathbb{R}^n$ gives the canonical 1-form $\sum_{i=1}^n y_i dx_i$.

□

—

$$T^*|\mathbb{R}^n = |\mathbb{R}^n \times (|\mathbb{R}^n)^*$$

$$= (\alpha_1, \dots, \alpha_n, y_1, \dots, y_n)$$

$$\pi : T^*|\mathbb{R}^n \rightarrow |\mathbb{R}^n$$

$$(\alpha_1, \dots, \alpha_n, y_1, \dots, y_n) \mapsto (\alpha_1, \dots, \alpha_n)$$

Definition 0.74. The map π is the **projective map** defined on the coordinates of $T^*|\mathbb{R}^n$.

Compute in the given coordinates the differential (derivative) of the projective map π :

$$d\pi : T(T^*|\mathbb{R}^n) \rightarrow T|\mathbb{R}^n$$

Let $v = (v_1, \dots, v_n, v_{n+1}, \dots, v_{2n}) \in T|\mathbb{R}^n \times (|\mathbb{R}^n)^*$.

$$d\pi_{(\alpha, \alpha)}(v) = (v_1, \dots, v_n)$$

(The projection onto the first n coordinates.)

$$\lambda_{(\alpha, \alpha)}(v) = d(d\pi_{(\alpha, \alpha)}(v))$$

$$= (\alpha_1, \alpha_2, \dots, \alpha_n) \cdot (v_1, \dots, v_n)$$

$$= \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n$$

Remark 0.21. The expression $(\alpha_1, \dots, \alpha_n)$ is the coordinate representation of the covector

$$d_1 d\alpha_1 + d_2 d\alpha_2 + \dots + d_n d\alpha_n$$

acting on $(v_1, \dots, v_n)$.

Remark 0.22. The expression for $v_1 \frac{\partial}{\partial \alpha_1} + v_2 \frac{\partial}{\partial \alpha_2} + \cdots + v_n \frac{\partial}{\partial \alpha_n}$ is the tangent vector representation of v .

—

Therefore, if we consider $(d\alpha_1, \dots, d\alpha_n) \cdot (v_1, \dots, v_n)$, we have

$$(d\alpha_1, \dots, d\alpha_n) \cdot (v_1, \dots, v_n) = \left(\sum_{i=1}^n \alpha_i d\alpha_i \right) \left(\sum_{i=1}^n v_i \frac{\partial}{\partial x_i} \right) = \sum_{i=1}^n \alpha_i v_i.$$

We write the 1-form λ as follows:

$$\lambda_{x,y} = \sum_{i=1}^n y_i d\alpha_i.$$

This implies that

$$d\lambda = \sum_{i=1}^n dy_i \wedge d\alpha_i.$$

We define a 2-form on T^*X by taking the exterior derivative of λ :

$$\omega = d\lambda.$$

Claim 8. The 2-form ω is a symplectic form in particular

- (i) ω is closed, i.e., $d^2\omega = 0$ is always true.
- (ii) ω is non-degenerate.

Now choose a chart with coordinates (q_i, p_i) and then ω can be seen as Ω^{2n} .

Exercise

Verify the claim 2

Hint: Use a coordinate chart at any pt (x, a) .

Example 0.23. Let $X = S^1$.

We have

$$T^*X = \{(\theta, v) \mid \theta \in S^1, v \in T_\theta^*S^1\}$$

and

$$T^*X = S^1 \times \mathbb{R}.$$

What is λ on $T^*X = S^1 \times \mathbb{R}$?

$$T(T^*X) = T(S^1 \times \mathbb{R}) = TS^1 \times \mathbb{R}.$$

Let (M, ω) be a symplectic manifold of dimension $2n$.

Let $H : M \rightarrow \mathbb{R}$ be a smooth function H , when fixed, will be referred to as a Hamiltonian function, or simply Hamiltonian.

Definition 0.75. Given a Hamiltonian function H as M , we get a unique vector field associated to H on M as follows: Consider the 1-form given by dH on M . Since ω is a non-degenerate 2-form, we get a unique vector field X_H defined by

$$\omega(X_H, \cdot) = dH(\cdot).$$

We call X_H the Hamiltonian vector field associated with it.

Take $(CM, dp \wedge dq)$

$[\mathbb{R}^{2n}$ is equipped with the coordinates $(q, p)]$

Remark 0.23. If ω is a symplectic form on M then $-\omega$ is also a symplectic form $(-\omega)(u, v) = -\omega(u, v)$.

Take the Hamiltonian function $H(q, p) = \frac{p^2}{2m}$ where $m > 0$ is a constant.

Compute the associated Hamiltonian vector field X_H .

PART 1 dH (= Total derivative of H)

$$dH = \frac{\partial H}{\partial q} dq + \frac{\partial H}{\partial p} dp$$

PART 2 Now find X_H such that

$$dp \wedge dq(X_H, \cdot) = \frac{\partial H}{\partial q} dq(\cdot) - \frac{\partial H}{\partial p} dp(\cdot)$$

Let $X_H = a(p, q) \frac{\partial}{\partial q} + b(p, q) \frac{\partial}{\partial p}$.

$$dp \wedge dq \left(a(p, q) \frac{\partial}{\partial q} + b(p, q) \frac{\partial}{\partial p}, \cdot \right) = \frac{p}{m} dp(\cdot)$$

Using linearity in the first variable, the left-hand side expands to give

$$a(p, q) dp \wedge dq \left(\frac{\partial}{\partial q}, \cdot \right) + b(p, q) dp \wedge dq \left(\frac{\partial}{\partial p}, \cdot \right)$$

$$a(p, q) \begin{vmatrix} 0 & dp(\cdot) \\ 1 & dq(\cdot) \end{vmatrix} + b(p, q) \begin{vmatrix} 1 & dp(\cdot) \\ 0 & dq(\cdot) \end{vmatrix}$$

$$= -a(p, q) dp(\cdot) + b(p, q) dq(\cdot)$$

Now

$$-a(p, q) dp + b(p, q) dq = \frac{p}{m} dp.$$

By comparing coefficients, we have

$$\begin{aligned} a(p, q) &= -\frac{p}{m}, \\ b(p, q) &= 0. \end{aligned}$$

Hence,

$$X_H = -\frac{p}{m} \frac{\partial}{\partial q}.$$

Plot the vector field on $\mathbb{R}^2$. (HW)

Example 0.24. Consider the Hamiltonian

$$H = \frac{1}{2}(p^2 + q^2)$$

defined on $(\mathbb{R}^2, dp \wedge dq)$.

The differential of H is given by

$$dH = q dq + p dp.$$

For the symplectic vector field X_H associated with H on $(\mathbb{R}^2, dp \wedge dq)$, if

$$X_H = a(p, q) \frac{\partial}{\partial q} + b(p, q) \frac{\partial}{\partial p},$$

then by the same process described above, we find

$$a(p, q) = -p, \quad b(p, q) = q.$$

Thus, the Hamiltonian vector field is

$$X_H = -p \frac{\partial}{\partial q} + q \frac{\partial}{\partial p}.$$

Definition 0.76. Let X be a smooth manifold of dimension n . A **vector field** V on X is a smooth assignment of a tangent vector at each point of X . In particular, $V: X \rightarrow TX$ is a smooth map such that

$$x \mapsto V(x) \in T_x X.$$

Definition 0.77. A **flow** on X is a 1-parameter family of diffeomorphisms $\{\varphi_t: X \rightarrow X\}_{t \in \mathbb{R}}$.

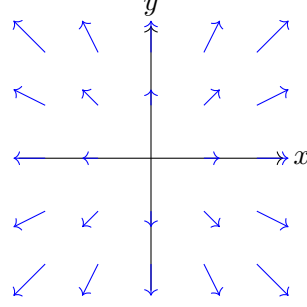
Theorem 0.8. Let V be a vector field on X . Then V generates a flow $\varphi_t: X \rightarrow X$ for $0 \leq t < \varepsilon$ such that

- (i) $\varphi_0 = \text{Id}_X$ (the identity diffeomorphism),

- (ii) $\varphi_{t_1+t_2} = \varphi_{t_1} \circ \varphi_{t_2}$,
- (iii) $\frac{d}{dt}\big|_{t=t_0} \varphi_t(\beta) = V(\varphi_{t_0}(\beta))$.

Example 0.25. Consider $\mathbb{R}^2$.

(1) $V_1(x, y) = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y}$.



V_1 : radial flow, $\varphi_t(x, y) = e^t(x, y)$

We now compute the flow of V_1 .

Let (x_0, y_0) be a fixed point in $\mathbb{R}^2$. The flow φ_t of V_1 satisfies (3) of theorem

$$\frac{d}{dt}\varphi_t(x_0, y_0) = V_1(\varphi_t(x_0, y_0)).$$

Here $\varphi_t : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a diffeomorphism for each t .

Let $(x(t), y(t))$ be the trajectory followed by (x_0, y_0) .

In other words, we have the following:

$$(\varphi_t(x_0, y_0)) = (x(t), y(t))$$

Therefore, we have

$$\frac{d}{dt}\varphi_t(x_0, y_0) = (\dot{x}(t), \dot{y}(t))$$

where $\dot{}$ represents the derivative with respect to t .

By property (3),

$$\begin{aligned} (\dot{x}(t), \dot{y}(t)) &= V_1(x(t), y(t)) \\ \implies (\dot{x}(t), \dot{y}(t)) &= x(t) \frac{\partial}{\partial x} + y(t) \frac{\partial}{\partial y} \end{aligned}$$

Therefore, we have a system of ordinary differential equations (ODEs) given by

$$\dot{x}(t) = x(t) \quad \text{and} \quad \dot{y}(t) = y(t)$$

with the initial condition $x(0) = x_0$ and $y(0) = y_0$.

Solving the above system of ODEs, we get

$$\begin{aligned}x(t) &= x_0 e^t, & y(t) &= y_0 e^t \\ \varphi_t(x_0, y_0) &= (x_0 e^t, y_0 e^t) = e^t(x_0, y_0)\end{aligned}$$

Exercise 21. Verify property (2):

$$\varphi_{t_1+t_2}(x_0, y_0) = (e^{t_1+t_2}x_0, e^{t_1+t_2}y_0) = e^{t_1}(e^{t_2}x_0, e^{t_2}y_0) = \varphi_{t_1} \circ \varphi_{t_2}(x_0, y_0)$$

2)

Let φ_t denote the flow of V_2 . Thus,

$$\varphi_t(x_0, y_0) = (x(t), y(t)) \quad \text{with} \quad x(0) = x_0 \quad \text{and} \quad y(0) = y_0.$$

Then,

$$\frac{d}{dt}\varphi_t(x_0, y_0) = V_2(\varphi_t(x_0, y_0))$$

implies

$$(\dot{x}(t), \dot{y}(t)) = (-y(t), x(t)).$$

Thus,

$$\dot{x}(t) = -y(t) \quad \text{and} \quad \dot{y}(t) = x(t).$$

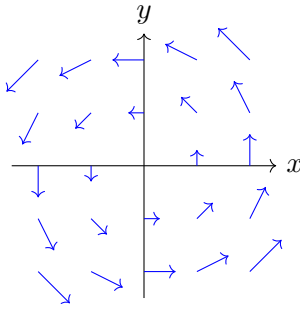
Differentiating again,

$$\ddot{x}(t) + x(t) = 0 \quad \text{and} \quad \ddot{y}(t) + y(t) = 0.$$

Solving the above system, we get

$$x(t) = x_0 \cos t - y_0 \sin t,$$

$$y(t) = x_0 \sin t + y_0 \cos t.$$



V_2 : rotational flow, $\varphi_t(x, y) = R_t(x, y)$

3)

Let $V_3(x, y) = y \frac{\partial}{\partial x}$.

$$\frac{d}{dt}\varphi_t(x_0, y_0) = V_3(\varphi_t(x_0, y_0)) = (y(t), 0)$$

implies

$$\dot{x}(t) = y(t) \quad \text{and} \quad \dot{y}(t) = 0.$$

Thus,

$$\begin{aligned} \ddot{x}(t) = 0 &\Rightarrow x(t) = y_0 t + x_0, \\ y(t) &= y_0. \end{aligned}$$

Therefore,

$$\varphi_t(x_0, y_0) = (y_0 t + x_0, y_0).$$

Theorem 0.9. Let X be a smooth manifold of dimension n . Assume that X is a compact manifold (without boundary). Then for every vector field V on X , the flow φ_t for V exists for all $t \in \mathbb{R}_{\geq 0}$.

Theorem 0.10. (This part works for non-compact manifolds)

If a vector field V on X has the property that the flow of V , if it exists, then it exists for ε -time around uniformly each point $x \in X$. Then the flow φ_t of V exists for all time $t \in \mathbb{R}$.

Note that we have an isomorphism

$$d\varphi_t: T_p M \rightarrow T_{\varphi_t(p)} M.$$

This motivates the following definition of the rate of change of β along the flow of V .

$$\lim_{t \rightarrow 0} \frac{\beta_{\varphi_t(p)}(d\varphi_t(\cdot), \dots, d\varphi_t(\cdot)) - \beta_p(\cdot, \dots, \cdot)}{t}$$

Definition 0.78. Let us denote

$$\beta_{\varphi_t(p)}(d\varphi_t(\cdot), \dots, d\varphi_t(\cdot))$$

by $\varphi_t^* \beta$. The form $\varphi_t^* \beta$ is called the pullback of β via φ_t .

The rate of change of β ,

$$\lim_{t \rightarrow 0} \frac{\varphi_t^* \beta - \beta}{t}.$$

—

Suppose a k -form β is given on X . We would like to find how β changes under the flow of a vector field V .

Definition 0.79. Let β be a k -form on X . The rate of change of β under the flow of V is defined as follows:

Let φ_t denote the flow of V . Then, the Lie derivative of β with respect to the vector field V , denoted $L_V \beta$, is given by

$$L_V \beta = \lim_{h \rightarrow 0} \frac{\varphi_h^* \beta - \beta}{h}.$$

This is called the Lie derivative of β with respect to the vector field V .

—

Example 0.26. Consider $X = \mathbb{R}^2$ with (x, y) coordinates. Define the vector field

$$V_1(x, y) = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y}.$$

Let $\varphi_t(x_0, y_0) = e^t(x_0, y_0)$ be the flow generated by V_1 . Take the 1-form

$$\beta_{x,y} = x dx + y dy.$$

The differential of φ_t is given by

$$d\varphi_t = \begin{bmatrix} e^t & 0 \\ 0 & e^t \end{bmatrix}.$$

Compute the Lie derivative $L_{V_1}\beta$. First, compute $\varphi_t^*\beta = \beta_{\varphi_t(\cdot)} \circ (d\varphi_t(\cdot))$:

$$\varphi_t^*\beta \left(\frac{\partial}{\partial x} \right) = \beta_{\varphi_t(x_0, y_0)} \left(d\varphi_t \left(\frac{\partial}{\partial x} \right) \right) = e^t \beta_{(x_0, y_0)} \left(\frac{\partial}{\partial x} \right).$$

Explicitly,

$$\varphi_t^*\beta \left(\frac{\partial}{\partial x} \right) = e^t (e^t x_0 dx + e^t y_0 dy) \left(\frac{\partial}{\partial x} \right) = e^{2t} x_0.$$

Similarly,

$$\varphi_t^*\beta \left(\frac{\partial}{\partial y} \right) = e^{2t} y_0.$$

$$\begin{aligned} L_{v_1} B \left(\frac{\partial}{\partial x} \right) &= \lim_{h \rightarrow 0} \frac{\varphi_{t+h}^* B - \varphi_t^* B}{h} \left(\frac{\partial}{\partial x} \right) \\ &= \lim_{h \rightarrow 0} \frac{e^{2t(h)} x_0 - e^{2t} x_0}{h} \\ &= \lim_{h \rightarrow 0} e^{2t} x_0 \left(\frac{e^{2h} - 1}{h} \right) \\ &= e^{2t} x_0 \cdot 2 \end{aligned}$$

$$L_{v_1} B \left(\frac{\partial}{\partial y} \right) = 2e^{2t} y_0$$

In particular, Lie derivatives at $t = 0$ is given by

$$L_{v_1} B \left(\frac{\partial}{\partial x} \right) \Big|_{t=0} = 2x_0$$

$$L_{v_1} B \left(\frac{\partial}{\partial y} \right) \Big|_{t=0} = 2y_0$$

$$L_{v_1}B = 2x_0 dx + 2y_0 dy.$$

Let M be a smooth manifold of dimension n .

Definition 0.80. Let X be a vector field on M . The **contraction** of a k -form β along X is defined as follows:

We define the map

$$i_X: \Omega^k \rightarrow \Omega^{k-1},$$

where for $\beta \in \Omega^k$,

$$i_X\beta \in \Omega^{k-1}.$$

For $\beta \in \Omega^k$, the contraction $i_X\beta$ is defined on the $(k-1)$ -fold product of tangent spaces by

$$i_X\beta: T_pM \times \cdots \times T_pM \rightarrow \mathbb{R},$$

where

$$i_X\beta(v_1, \dots, v_{k-1}) = \beta(X, v_1, \dots, v_{k-1}).$$

Example 0.27. Consider $\mathbb{R}^2$ with coordinates (x, y) . Let $\omega = dx \wedge dy$ and $X = y \frac{\partial}{\partial x}$. We compute $i_X\omega$.

For $v \in T_p\mathbb{R}^2$,

$$i_X\omega(v) = \omega(X, v) = \omega\left(y \frac{\partial}{\partial x}, v\right).$$

—

$$= dx \wedge dy\left(y \frac{\partial}{\partial x}, \nu\right) = c_1 dx + c_2 dy$$

Now, for $\nu = e_1 = (1, 0)$, $e_2 = (0, 1)$,

$$dx \wedge dy\left(y \frac{\partial}{\partial x}, \frac{\partial}{\partial x}\right) dx + dx \wedge dy\left(y \frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right) dy$$

$$= \begin{vmatrix} y & 1 \\ 0 & 0 \end{vmatrix} dx + \begin{vmatrix} y & 0 \\ 0 & 1 \end{vmatrix} dy$$

$$= y dy$$

Now take

$$X = -y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}$$

By definition,

$$i_X(dx \wedge dy) = c_1 dx + c_2 dy$$

where

$$c_1 = dx \wedge dy \left(X, \frac{\partial}{\partial x} \right)$$

$$c_2 = dx \wedge dy \left(X, \frac{\partial}{\partial y} \right)$$

Proof.

$$i_X(dx \wedge dy) = -x dx - y dy$$

□

Cartan's magic formula states the following:

$$\mathcal{L}_X \beta = d(i_X \beta) + i_X(d\beta)$$

for any k -form β on M .

Apply Cartan's Magic Formula to compute $\mathcal{L}_{y \frac{\partial}{\partial x}}$:

$$\mathcal{L}_{y \frac{\partial}{\partial x}}(dx \wedge dy) = d \left(y \frac{\partial}{\partial x} \lrcorner (dx \wedge dy) \right) + y \frac{\partial}{\partial x} \lrcorner d(dx \wedge dy)$$

$$= d(y dy) = dy \wedge dy = 0.$$

Exercise: Cartan's Magic Formula on all of the Lie derivatives so far.

Compute $\mathcal{L}_{\frac{\partial}{\partial x}}(dx \wedge dy)$

Compute $\mathcal{L}_{\frac{\partial}{\partial x}}(\sum_{i=1}^n dx_i \wedge dy_i)$ (Use the fact that i_X is a linear operator)

Compute $\mathcal{L}_{\frac{\partial}{\partial x_i}}(dx_1 \wedge dy_1 \wedge \cdots \wedge dy_n)$

$$d(dx_1 \wedge \cdots \wedge dy_n) = 0$$

$$i_{\frac{\partial}{\partial x_i}}(dx_1 \wedge \cdots \wedge dy_n) = dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dy_n \quad (\partial/\partial x_i, \dots)$$

The only non-zero

$$\left| \frac{\partial y_1}{\partial x_1}, \dots, \frac{\partial y_n}{\partial x_n} \right| = 1$$

and $d(\bar{I}_{x_1 \dots x_n}(dx_1 \wedge \cdots \wedge dx_n)) = 0$.

Proposition 0.3. Let M be a smooth manifold, for a k -form β and vector field X on M . Then, the flow φ_t of X keeps β invariant. In other words, $\varphi_t^* \beta = \beta \forall t$.

Theorem 0.11 (Liouville's Theorem (stronger)). 1. Let (M, ω) be a symplectic manifold of dimension $2n$.

2. Let $H : M \rightarrow \mathbb{R}$ be a smooth function.

Then, the flow associated to the Hamiltonian vector field X_H keeps ω invariant, i.e., if φ_t denotes the flow of X_H , then

$$\varphi_t^* \omega = \omega \quad \forall t.$$

—

Proof. It is sufficient (in light of the prop) to show

$$\mathcal{L}_{X_H} \omega = 0.$$

By Cartan's Magic formula

$$\begin{aligned} \mathcal{L}_{X_H} \omega &= di_{X_H} \omega + i_{X_H} d\omega \\ \implies \mathcal{L}_{X_H} \omega &= di_{X_H} \omega. \end{aligned}$$

Recall that $i_{X_H} \omega = \omega(X_H, \cdot) = dH(\cdot)$ by defining property of X_H

$$\implies \mathcal{L}_{X_H} \omega = d(i_{X_H} \omega) = d(dH) = 0.$$

□

Example 0.28. Illustration

Take $M = \mathbb{R}^2$, $\omega = dp \wedge dq$ on

$$T^*(\mathbb{R}^2) = \{(q, p) \mid p, q \in \mathbb{R}^2\}$$

and consider

$$H(p, q) = \frac{1}{2}(p^2 + q^2).$$

Then

$$dH = p \, dp + q \, dq.$$

—

Therefore, the Hamiltonian vector field is given by

$$X_H = -p \frac{\partial}{\partial q} + q \frac{\partial}{\partial p}.$$

The flow of the Hamiltonian system is given by

$$\varphi_t(q, p) = (q \cos t - p \sin t, q \sin t + p \cos t).$$

Let $M = S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$.

Consider the symplectic form given by the area form

$$\omega$$

for any $u, v \in T_p S^2$,

$$\omega(u, v) = (u \times v) \cdot p.$$

Take $H : S^2 \rightarrow \mathbb{R}$ defined by

$$H(x, y, z) = z.$$

Compute:

1. $dH = dz \in \Omega^1(S^2)$
2. $X_H = ?$
3. $\varphi_t = ?$

By the given condition, we have

$$\omega(X_h, \cdot) = dH(\cdot).$$

$$\omega(x_h|_{(c_0,0,1)}, \cdot) = dz|_{(c_0,0,1)} = 0.$$

By non-degeneracy, $X_h|_{(c_0,0,1)} = 0$.

Suppose $p = (x_0, y_0, z_0)$.

Then

$$T_p S^2 = \{(u, v, w) \mid (u, v, w) \cdot (x_0, y_0, z_0) = 0\}.$$

Now, $dz : T_p S^2 \rightarrow \mathbb{R}$ is defined by

$$(u, v, w) \mapsto w.$$

$$dz(u dx + v dy + w dz) = w.$$

Therefore,

$$\omega(X_h, u dx + v dy + w dz) = dz(u dx + v dy + w dz) = w.$$

$$(X_h \times (u dx + v dy + w dz)) \cdot (x_0 dx + y_0 dy + z_0 dz) = w.$$

Let $X_h = c_1 dx + c_2 dy + c_3 dz$, then

$$X_h \times (u dx + v dy + w dz) = (c_2 w - c_3 v) dx - (c_1 w - c_3 u) dy + (c_1 v - c_2 u) dz.$$

—

We have the following equation satisfied by all $(u, v, w) \in T_p S^2$:

$$(X_h \times (u dx + v dy + w dz)) \cdot (x_0 dx + y_0 dy + z_0 dz) = w.$$

This simplifies to:

$$x_0(c_2 w - c_3 v) - y_0(c_1 w - c_3 u) + z_0(c_1 v - c_2 u) = w.$$

The equation is satisfied by all $(u, v, w) \in T_p S^2$. Take $(u, v, w) = (-y_0, x_0, 0)$ as it satisfies $(*)$.

Substituting the above choice, we get:

$$x_0(-c_3x_0) - y_0(c_1 \cdot 0 - c_3(-y_0)) + z_0(c_1x_0 + c_2y_0) = 0.$$

Simplifying further:

$$-c_3x_0^2 - c_3y_0^2 + z_0(c_1x_0 + c_2y_0) = 0.$$

We then obtain three equations and solve for c_1, c_2, c_3 .

Given a symplectic form (M, ω) , we have a natural choice of volume form, i.e. $2n$ -form, given by

$$\omega^n = \underbrace{\omega \wedge \omega \wedge \cdots \wedge \omega}_{n \text{ form wedge product}}.$$

Exercise 22. On $\mathbb{R}^4$ consider $\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$ with coordinate system (q_1, q_2, p_1, p_2) .

Here $\omega^2 = (dq_1 \wedge dp_1 + dq_2 \wedge dp_2)^2$

$$= dq_1 \wedge dp_1 \wedge dq_1 \wedge dp_1 + dq_2 \wedge dp_2 \wedge dq_2 \wedge dp_2$$

$$+ dq_1 \wedge dp_1 \wedge dq_2 \wedge dp_2 + dq_2 \wedge dp_2 \wedge dq_1 \wedge dp_1.$$

Now $dq_2 \wedge dp_2 \wedge dq_1 \wedge dp_1 = dq_1 \wedge dp_1 \wedge dq_2 \wedge dp_2$.

Hence

$$\omega^2 = 2 \cdot dq_1 \wedge dp_1 \wedge dq_2 \wedge dp_2.$$

In general on $\mathbb{R}^{2n}$ with coordinates $(q_1, \dots, q_n, p_1, \dots, p_n)$ and $\omega = \sum_{i=1}^n dq_i \wedge dp_i$.

Compute:

$$\omega^n = \left(\sum_{i=1}^n dq_i \wedge dp_i \right)^n = n! \cdot (dq_1 \wedge dp_1) \wedge \cdots \wedge (dq_n \wedge dp_n)$$

In general, the $2n$ -form ω^n does not vanish at any point of M .

$$\omega^n : T_p M \times \cdots \times T_p M \rightarrow \mathbb{R}.$$

Suppose at p , $\omega^n = 0$.

$$\omega^n(v_1, \dots, v_{2n}) = 0 \quad \forall v_i \in T_p M.$$

Take a symplectic basis of a symplectic vector space. $(T_p M, \omega_p)$ denote it by $(e_1, \dots, e_n, f_1, \dots, f_n)$,

$$\omega_p(e_i, e_j) = \omega(f_i, f_j) = 0,$$

$$\omega(e_i, f_j) = \delta_{ij}.$$

$$(\omega \wedge \dots \wedge \omega)(e_1, \dots, e_n, f_1, \dots, f_n) \neq 0.$$

(Verify the computation.)

Therefore, w^n at any point p is a non-zero $2n$ -form.

On symplectic manifold (M, ω) , the $2n$ -form w^n gives us a volume.

Theorem 0.12 (Liouville's Theorem (Weak form)). The setup is the same as strong form. Then the flow $\varphi_t^{X_H}$ associated to H preserves the volume form w^n . In other words,

$$\mathcal{L}_{X_H} w^n = 0.$$

Proof. Strong form $\implies$ Weak form

$$\varphi_t^* w = w \quad \text{for all } t$$

Observe

$$\varphi^*(w \wedge \dots \wedge w) = (\varphi^* w)^n$$

□

Consider $\mathbb{R}^2$ with q, p_1 coordinates.

Let $H : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a smooth function

Definition 0.81. The Hamiltonian vector field X_H associated with H is defined by

$$X_H = \frac{\partial H}{\partial p_1} \frac{\partial}{\partial q} - \frac{\partial H}{\partial q} \frac{\partial}{\partial p_1}.$$

Theorem 0.13. If A is a region in $\mathbb{R}^2$, then the area of A is preserved under the flow of the Hamiltonian vector field X_H .

Proof. The flow of the Hamiltonian vector field X_H is given by the Hamiltonian system of differential equations

$$\frac{dq}{dt} = \frac{\partial H}{\partial p_1}, \quad \frac{dp_1}{dt} = -\frac{\partial H}{\partial q}.$$

The divergence of X_H is

$$\nabla \cdot X_H = \frac{\partial}{\partial q} \left(\frac{\partial H}{\partial p_1} \right) + \frac{\partial}{\partial p_1} \left(-\frac{\partial H}{\partial q} \right) = 0,$$

since H is smooth. By Liouville's theorem, the flow preserves volume (and hence area in $\mathbb{R}^2$). □